SEMNEST: proofs of the nesting and equivalence relations

all 80 items of the capsule

Generated with the capsule's item bank (`bank.js`) and checked in exact arithmetic (`.dev/proofs/build.js`); see also `.dev/verify_bank.js` and `.dev/verify_bank.R`

1 Setting and definitions

Each item compares two models, A (left) and B (right), for the same p observed variables. A model is written in RAM form

$$v = \mathbf{A}v + u, \qquad \mathrm{Cov}(u) = \mathbf{S}, \qquad \Sigma_{\mathrm{full}} = (I - \mathbf{A})^{-1}\mathbf{S}(I - \mathbf{A})^{-\top},$$

with v the observed and latent variables and Σ the observed block of Σ_{full}. $\mathbf{A}$ holds the drawn directed effects (loadings λ, regressions β, second-order loadings γ); $\mathbf{S}$ holds a free variance for every variable (the residual variance θ_{ii} or ψ of an endogenous variable) and the drawn covariances. For variables in a *network block*, the block of $\mathbf{S}$ is K^{-1}, where the precision matrix K has a free diagonal and free off-diagonal elements only for the drawn undirected edges: a Gaussian graphical model (GGM) on observed variables, a residual network model (RNM) on the residuals of indicators, or a latent network model (LNM) on latent variables (Epskamp, Rhemtulla & Borsboom, 2017, *Psychometrika*).

Parameters. The free parameters $\theta \in \mathbb{R}^q$ are unrestricted real numbers: variances may be negative (Heywood solutions), as in lavaan's default estimation. The scale of each latent variable is fixed by its first loading (marker variable); q_X counts the free parameters of model X. Let $\Theta_X = \{\theta : \Sigma_X(\theta) \text{ exists and is positive definite}\}$ (an open set) and

$$\mathcal{M}_X = \{\Sigma_X(\theta) : \theta \in \Theta_X\},$$

the set of covariance matrices model X can produce. $\overline{\mathcal{M}}$ denotes the closure within the positive definite matrices.

Definition. X is *nested* in Y, written $X \preceq Y$, if $\mathcal{M}_X \subseteq \overline{\mathcal{M}_Y}$: every covariance matrix that X can produce, Y can produce or approximate arbitrarily closely, so fitting Y to it gives a perfect fit ($F_{\mathrm{ML}} = 0$, $\chi^2 = 0$). This is exactly what the numerical check of Bentler & Satorra (2010) and of the PL5225 assignment tests. X and Y are *equivalent* ($X \equiv Y$) if $X \preceq Y$ and $Y \preceq X$. *A is nested in B* in the game means $A \preceq B$ and $B \not\preceq A$; *neither* means $A \not\preceq B$ and $B \not\preceq A$.

The closure makes the definition insensitive to the measure-zero exceptions that are typical for latent variable models (e.g. a mapping that divides by a covariance that happens to be zero). Allowing negative variances is what makes the textbook statements ("a one-factor model with three indicators is saturated", "a second-order factor on three factors is equivalent to three correlated factors") literally true.

2 Lemmas

Lemma (R, restriction)**.** *If X is obtained from Y by fixing free parameters of Y to constants or by equating them, then $X \preceq Y$.*

Proof. Θ_X embeds in Θ_Y with the same implied Σ. □

Lemma (G, generic maps)**.** *If there are a dense subset $D \subseteq \Theta_X$ and a map $\varphi : D \to \Theta_Y$ with $\Sigma_Y(\varphi(\theta)) = \Sigma_X(\theta)$ for all $\theta \in D$, then $X \preceq Y$.*

Proof. Σ_X is continuous, so $\mathcal{M}_X = \Sigma_X(\Theta_X) \subseteq \overline{\Sigma_X(D)} \subseteq \overline{\mathcal{M}_Y}$. In the items D is Θ_X minus the zero set of a polynomial that is not identically zero ("generic" parameter values), which is dense. □

Lemma (S, separating constraint)**.** *If a polynomial f in the entries of Σ vanishes on $\mathcal{M}_Y$ and $f(\Sigma_X(\theta^*)) \neq 0$ for some $\theta^* \in \Theta_X$, then $X \not\preceq Y$.*

Proof. f is continuous, so it vanishes on $\overline{\mathcal{M}_Y}$, which therefore does not contain $\Sigma_X(\theta^*)$. □

In every application below, f vanishing on $\mathcal{M}_Y$ is shown algebraically in the text (and was also checked in exact rational arithmetic at random rational parameter values), and the point θ^* with rational coordinates is listed in Appendix B together with the exact value of f.

Lemma (D, dimension)**.** *Let $J_X(\theta)$ be the Jacobian of $\theta \mapsto \operatorname{vech}\Sigma_X(\theta)$. If $\operatorname{rank} J_X(\theta^*) > q_Y$ for some $\theta^* \in \Theta_X$, then $X \not\preceq Y$.*

Proof. $\mathcal{M}_Y$ is the image of a semialgebraic set of dimension q_Y under a rational map, so $\overline{\mathcal{M}_Y}$ is semialgebraic with dimension at most q_Y. The rank of J_X is lower semicontinuous, so its maximum $r \geq \operatorname{rank} J_X(\theta^*) > q_Y$ is attained on a nonempty open subset of Θ_X, where the rank is constant; by the constant rank theorem $\mathcal{M}_X$ contains an r-dimensional embedded submanifold, which cannot lie in a semialgebraic set of dimension $< r$. □

The ranks were computed exactly (rational arithmetic) at the rational points listed in Appendix B.

Lemma (B, saturated blocks)**.** *(a) The following are saturated ($\overline{\mathcal{M}}$ = all positive definite matrices): the model with all covariances free; a complete recursive path model (every pair connected by a directed path following a fixed order); a one-factor model with three indicators. (b) If two models differ only in how they model the covariance matrix of a block of variables that is connected to the rest of the model in the same way, and both can produce every covariance matrix of that block (up to closure), the models are equivalent; if only one of them can, the other is nested in it.*

Proof. (a) Every $\Sigma \succ 0$ has a Cholesky factorization $\Sigma = (I-B)^{-1}D(I-B)^{-\top}$ with B strictly lower triangular and D diagonal; that is a complete recursive path model. For one factor with three indicators and $\sigma_{12}\sigma_{13}\sigma_{23} \neq 0$: $\lambda_1 = 1$, $\psi = \sigma_{12}\sigma_{13}/\sigma_{23}$, $\lambda_2 = \sigma_{23}/\sigma_{13}$, $\lambda_3 = \sigma_{23}/\sigma_{12}$, $\theta_{ii} = \sigma_{ii} - \lambda_i^2\psi$ reproduce Σ; apply Lemma G. (b) The implied Σ depends on the block only through the block's covariance matrix (e.g. $\Sigma = \Lambda\Psi\Lambda^\top + \Theta$ for the latent block). □

Lemma (M, merging factors)**.** *In a factor model where the covariance matrix Ψ of the factors is free, setting the variances and covariances of factors F_k and F_l equal ($\psi_{kk} = \psi_{ll} = \psi_{kl}$ and $\psi_{km} = \psi_{lm}$ for all other m) makes $F_k = F_l$ almost surely. Hence the model in which F_k and F_l are replaced by one factor loading on the union of their indicators is nested in it.*

Proof. With these values $\operatorname{Var}(F_k - F_l) = 0$; the implied Σ is that of the merged model (Lemma R on the parameters). □

Lemma (P, pairs)**.** *Let F be a latent variable of model X and y_i, y_j two observed variables whose only paths are loadings from F. Let X^+ be X plus the residual covariance $y_i \leftrightarrow y_j$, and X^G be X in which y_i, y_j instead load on a new latent variable G whose only path is $F \to G$. Then $X^+ \equiv X^G$.*

Proof. In X^G write $G = \gamma F + u$ with u independent of all other variables. Then $y_i = \lambda_i^G\gamma F + \lambda_i^G u + e_i$, and likewise for y_j: this is X^+ with loadings $\lambda_i = \lambda_i^G\gamma$, $\lambda_j = \lambda_j^G\gamma$, residual covariance $\theta_{ij} = \lambda_i^G\lambda_j^G\operatorname{Var}(u)$ and residual variances $\theta_{ii} + (\lambda_i^G)^2\operatorname{Var}(u)$; nothing else changes. Conversely, given X^+ with $\lambda_i\lambda_j \neq 0$, take $\gamma = 1$, $\lambda_i^G = \lambda_i$, $\lambda_j^G = \lambda_j$, $\operatorname{Var}(u) = \theta_{ij}/(\lambda_i\lambda_j)$ (possibly negative) and residual variances $\theta_{ii} - \lambda_i^2\operatorname{Var}(u)$. Lemma G gives both directions. □

Note that if F is exogenous and G has no other paths, $F \to G$ and $F \leftrightarrow G$ give the same (saturated) 2×2 latent block, so by Lemma B(b) G may also simply be a factor correlated with F.

Lemma (I, inverses)**.** *For a positive definite Σ, $(\Sigma^{-1})_{ij} = 0$ iff the partial covariance of i and j given all other variables is 0. Σ is block-diagonal iff Σ^{-1} is block-diagonal (with the same blocks). For three variables, $(\Sigma^{-1})_{13} = 0$ iff $\sigma_{12}\sigma_{23} - \sigma_{13}\sigma_{22} = 0$.*

Proof. Standard: $(\Sigma^{-1})_{ij} = C_{ji}/\det\Sigma$ with C the cofactor matrix; the inverse of a block-diagonal matrix is block-diagonal; for 3×3, $C_{31} = \sigma_{12}\sigma_{23} - \sigma_{13}\sigma_{22}$. □

Lemma (C, trees)**.** *Let a path model be a DAG whose skeleton is a tree and in which no variable has two parents (no colliders: chains and forks). Then its set of covariance matrices equals the GGM on the skeleton, $\{\Sigma \succ 0 : (\Sigma^{-1})_{ij} = 0$ for every non-adjacent pair$\}$.*

Proof. For a DAG, $\Sigma^{-1} = (I-B)^\top D^{-1}(I-B)$ with B supported on the arrows. If every node has at most one parent, $\big((I-B)^\top D^{-1}(I-B)\big)_{ij}$ for $i \neq j$ is nonzero only if i is the parent of j or vice versa (a product term $b_{ki}b_{kj}$ would need a common child k with two parents). So the DAG is contained in the GGM. Conversely, a $\Sigma \succ 0$ with tree-shaped inverse factorizes along any orientation of the tree away from a root (the Gaussian density factorizes over a tree as $\prod_k p(x_k \mid x_{\mathrm{pa}(k)})$; see Lauritzen, 1996), and every collider-free orientation of a tree points away from some root. □

Lemma (T, perfect indicators)**.** *In a factor model, an indicator y_1 of F with $\theta_{11} = 0$ and $\lambda_1 = 1$ equals F. Hence the path model in which y_1 is an exogenous cause of the other indicators of F ($y_k = \lambda_k y_1 + e_k$) is nested in the factor model.*

Proof. Take $\theta_{11} = 0$ (allowed: parameters are unrestricted) and $\psi = \mathrm{Var}(y_1)$; Lemma R. □

Lemma (H, higher-order and bifactor; Yung, Thissen & McLeod, 1999)**.** *A higher-order model (first-order factors F_k with disjoint indicator sets, $F_k = \gamma_k G + \zeta_k$) is nested in the bifactor model with a general factor on all indicators and one orthogonal group factor per first-order factor.*

Proof. For an indicator y_i of F_k: $y_i = \lambda_i\gamma_k G + \lambda_i\zeta_k + e_i$. The general factor G and the disturbances ζ_k are mutually uncorrelated, so this is a bifactor model with general loadings $\lambda_i\gamma_k$, group factor $S_k = \zeta_k$ with loadings λ_i and variance $\mathrm{Var}(\zeta_k)$ (possibly negative). Rescale G to its marker loading (generic: $\lambda_1\gamma_1 \neq 0$). □

3 How to read an item page

Each page shows the two path diagrams exactly as in the game (observed variables as squares, latent variables as circles, residuals as small circles, covariances as curved double-headed arrows, network edges as green lines; variances are free and not drawn), the model syntax ($\sim\sim$ is a covariance, `-` a network edge, `network` lists a network block, `a*` an equality label), the claim and its proof. (i) concerns $A \preceq B$, (ii) concerns $B \preceq A$. σ_{ij} is the covariance of y_i and y_j.

Contents

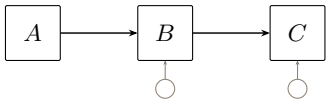
E01 Chain versus fork . . . 4
E02 A chain and its reverse . . . 5
E03 Regression or correlation between two factors . . . 6
E04 Two saturated models . . . 7
E05 One factor with three indicators is saturated . . . 8
E06 A cross-loading . . . 9
E07 One factor versus two . . . 10
E08 Adding a direct effect . . . 11
E09 A latent chain versus correlated factors . . . 12
E10 Orthogonal versus correlated factors . . . 13
E11 One residual covariance more . . . 14
E12 Correlated or independent predictors . . . 15
E13 Equal loadings . . . 16
E14 A direct effect of the covariate . . . 17
E15 The independence model . . . 18
E16 Chain versus collider . . . 19
E17 Two different residual covariances . . . 20
E18 Two orthogonal factors versus one factor . . . 21
E19 Two different mediators . . . 22
E20 Latent chain versus latent collider . . . 23
M01 Residual covariance versus a second factor . . . 24
M02 Higher-order model with three first-order factors . . . 25
M03 A two-indicator factor . . . 26
M04 The replacing rule . . . 27
M05 A saturated structural block . . . 28
M06 Higher-order model with four first-order factors . . . 29
M07 Two factors inside one factor with correlated residuals . . . 30
M08 Merging two factors . . . 31
M09 Every model is nested in a saturated model . . . 32
M10 Equal loading patterns . . . 33
M11 A factor model inside a saturated path model . . . 34
M12 Full versus partial mediation . . . 35

M13 A residual covariance as a factor . . . 36
M14 Dropping one path . . . 37
M15 A chain inside a saturated factor model . . . 38
M16 Cross-loadings on different items . . . 39
M17 Different splits of the items . . . 40
M18 Residual covariance versus cross-loading . . . 41
M19 Parallel versus serial mediation . . . 42
M20 Residual covariance on different pairs . . . 43
H01 Equivalent residual covariances . . . 44
H02 A counting trap . . . 45
H03 A three-indicator group factor equals a free residual block . . . 46
H04 One factor with paired residuals equals a higher-order model . . . 47
H05 Covariate or indicator . . . 48
H06 Higher-order inside bifactor . . . 49
H07 Correlated factors inside bifactor . . . 50
H08 Arrows the other way, still nested . . . 51
H09 A perfect indicator . . . 52
H10 Paired residuals inside three factors . . . 53
H11 Paired residuals inside four correlated factors . . . 54
H12 A covariate inside a factor model with one residual covariance . . . 55
H13 Replacing rule plus one restriction . . . 56
H14 A two-indicator factor inside residual pairs . . . 57
H15 Merging non-adjacent factors . . . 58
H16 Markov chain versus one factor . . . 59
H17 An illegal replacement . . . 60
H18 Higher-order versus one factor with one residual block . . . 61
H19 Two mediation chains . . . 62
H20 General factor versus latent chain . . . 63
P01 A path network is a chain . . . 64
P02 Orthogonal factors and network cliques . . . 65
P03 Latent network chain equals latent chain . . . 66
P04 Bifactor with two-indicator groups . . . 67
P05 Residual clique equals residual covariance block . . . 68
P06 Star network inside one factor . . . 69
P07 Orthogonal factors inside network cliques . . . 70
P08 Latent network chain inside a higher-order model . . . 71
P09 Isolated residual edges . . . 72
P10 A DAG inside its moral graph . . . 73
P11 One residual edge inside a residual chain . . . 74
P12 A chain inside a cycle network . . . 75
P13 A tree network inside two factors . . . 76
P14 A latent network inside a bifactor model . . . 77
P15 Factor plus complete residual network is saturated . . . 78
P16 Cycle network versus one factor . . . 79
P17 Residual chain network versus residual covariance chain . . . 80
P18 Correlated factors versus network cliques . . . 81
P19 Undirected path versus collider . . . 82
P20 Latent cycle network versus general factor . . . 83

E01 Chain versus fork

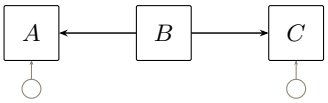
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 5$




```
A ~ B; C ~ B
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Both graphs have the skeleton $A - B - C$ and no collider, so by Lemma C both models equal the GGM with $(\Sigma^{-1})_{AC} = 0$, i.e. $\mathcal{M} = \{\Sigma \succ 0 : \sigma_{AC}\sigma_{BB} = \sigma_{AB}\sigma_{BC}\}$.

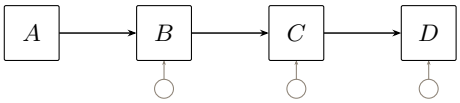
(ii) $B \preceq A$. See the previous line: $\mathcal{M}_A = \mathcal{M}_B$. □

In the game: Both models say that A and C are unrelated once you know B. Turning the first arrow around does not change that, so both reproduce exactly the same covariance matrices.

E02 A chain and its reverse

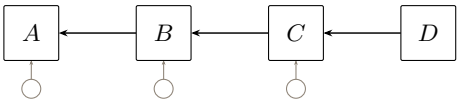
Level: Easy. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
B ~ A; C ~ B; D ~ C
```

Model B $q_B = 7$




```
C ~ D; B ~ C; A ~ B
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Both are chains on the path $A - B - C - D$ without colliders. By Lemma C both equal the GGM on that path: all $\Sigma \succ 0$ whose inverse is tridiagonal.

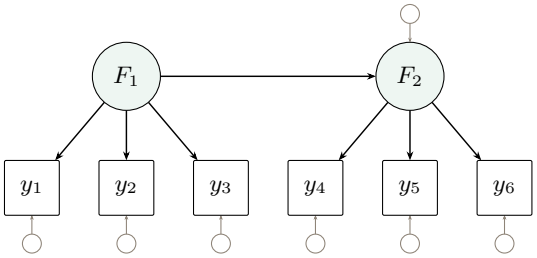
(ii) $B \preceq A$. Same argument: $\mathcal{M}_A = \mathcal{M}_B$. □

In the game: A chain read backwards is still a chain: in both models every variable only "talks" to its neighbours. Same conditional independencies, same fit on every dataset.

E03 Regression or correlation between two factors

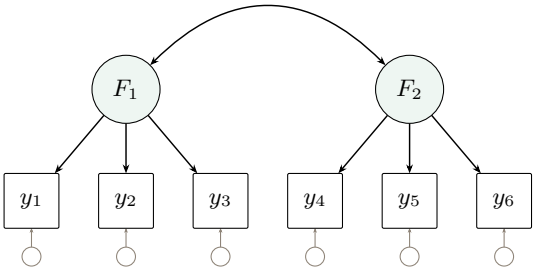
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F2 ~ F1
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. The measurement parts are identical; the models differ only in the latent block (F_1, F_2). In A, $\Psi = \begin{pmatrix} \psi_{11} & \beta\psi_{11} \\ \beta\psi_{11} & \beta^2\psi_{11} + \zeta \end{pmatrix}$; in B, Ψ is free. Both reach every 2×2 Ψ with $\psi_{11} \neq 0$ ($\beta = \psi_{12}/\psi_{11}$, $\zeta = \psi_{22} - \psi_{12}^2/\psi_{11}$), so by Lemmas B and G the models are equivalent.

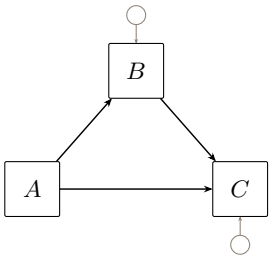
(ii) $B \preceq A$. Lemma B: in B the block is free, so $\mathcal{M}_A \subseteq \mathcal{M}_B$ trivially; with the map above $\mathcal{M}_B \subseteq \overline{\mathcal{M}_A}$. □

In the game: With only two factors, a regression F1 → F2 and a correlation F1 ↔ F2 both spend one parameter on the single covariance between the factors. The fit is identical.

E04 Two saturated models

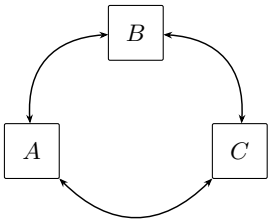
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 6$




```
B ~ A; C ~ A + B
```

Model B $q_B = 6$




```
A ~~ B + C; B ~~ C
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. A is a complete recursive path model and B the model with all covariances free; both are saturated (Lemma B(a)): $\overline{\mathcal{M}_A} = \overline{\mathcal{M}_B}$ = all positive definite 3×3 matrices.

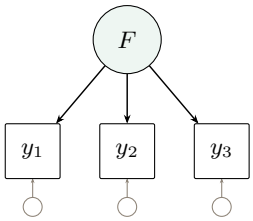
(ii) $B \preceq A$. As above. □

In the game: Both models are saturated: they have as many parameters (6) as there are variances and covariances. Every saturated model reproduces any covariance matrix, so all saturated models are equivalent.

E05 One factor with three indicators is saturated

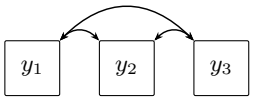
Level: Easy. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 6$




```
F =~ y1 + y2 + y3
```

Model B $q_B = 6$




```
y1 ~~ y2 + y3; y2 ~~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. B (all covariances free) is saturated, so $A \preceq B$.

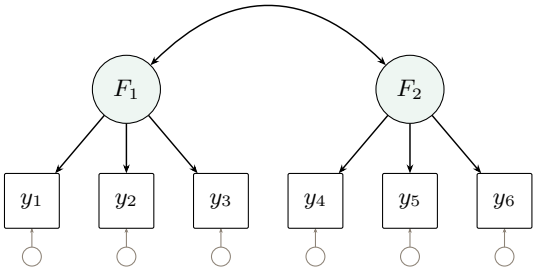
(ii) $B \preceq A$. By Lemma B(a) the one-factor model with three indicators reproduces every $\Sigma \succ 0$ with $\sigma_{12}\sigma_{13}\sigma_{23} \neq 0$ (a dense set): $\lambda_1 = 1$, $\psi = \sigma_{12}\sigma_{13}/\sigma_{23}$, $\lambda_2 = \sigma_{23}/\sigma_{13}$, $\lambda_3 = \sigma_{23}/\sigma_{12}$, $\theta_{ii} = \sigma_{ii} - \lambda_i^2\psi$. ($\psi < 0$ when the product of the covariances is negative: a Heywood solution.) Hence $B \preceq A$. □

In the game: A one-factor model with three indicators has 3 loadings and 3 residual variances: 6 parameters for 6 (co)variances. It is just-identified (saturated), like the model with all covariances free.

E06 A cross-loading

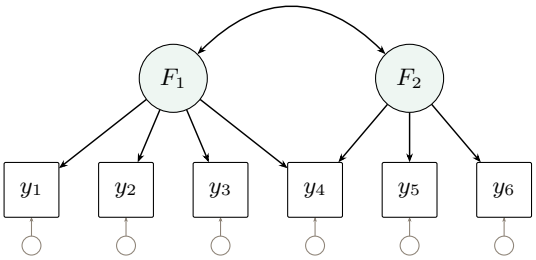
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix the cross-loading of y_4 on F_1 in B to 0 (Lemma R).

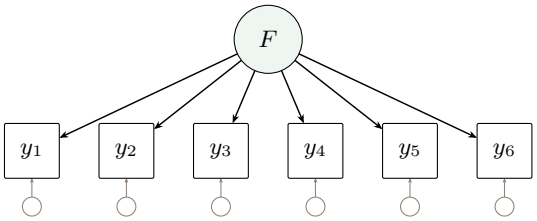
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 14 at the rational point listed in Appendix B, while model A has only $q_A = 13$ free parameters. □

In the game: B is A plus one cross-loading (F1 → y4). Fix that loading to zero and you get A back, so A is nested in B.

E07 One factor versus two

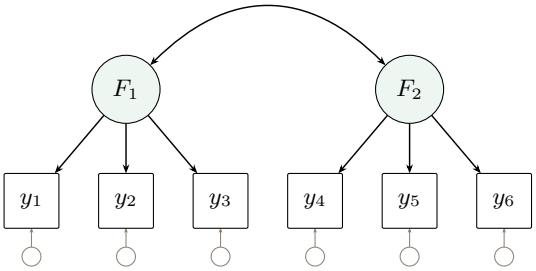
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma M: in B take $\psi_{11} = \psi_{22} = \psi_{12}$; then F_1 and F_2 act as one factor and B reproduces A.

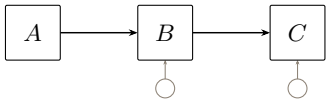
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 13 at the rational point listed in Appendix B, while model A has only $q_A = 12$ free parameters. □

In the game: A one-factor model is a two-factor model whose factors correlate perfectly (r = 1). So the one-factor model is nested in the two-factor model.

E08 Adding a direct effect

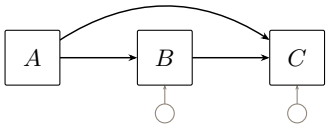
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 6$




```
B ~ A; C ~ A + B
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix $\beta_{CA} = 0$ in B (Lemma R).

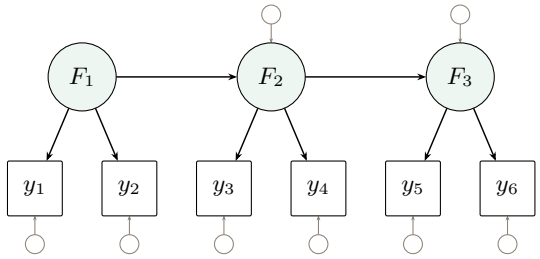
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 6 at the rational point listed in Appendix B, while model A has only $q_A = 5$ free parameters. □

In the game: B adds the direct effect A → C (and is saturated). Setting that effect to zero gives A: A is nested in B.

E09 A latent chain versus correlated factors

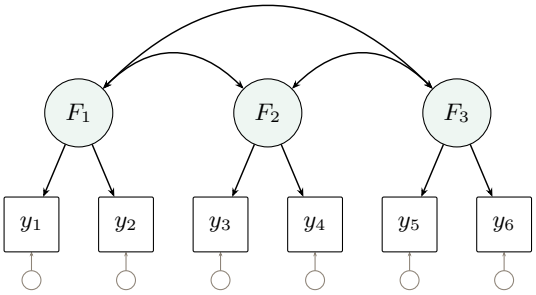
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Same measurement part; in B the latent covariance matrix Ψ is free (a saturated latent block), and the chain in A produces one particular Ψ. By Lemma B(b), $A \preceq B$.

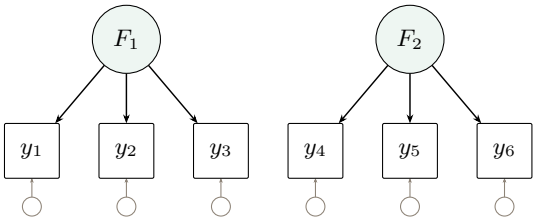
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: B leaves the three factor covariances free (a saturated latent block). The chain F1 → F2 → F3 restricts them (no direct F1–F3 link), so A is nested in B.

E10 Orthogonal versus correlated factors

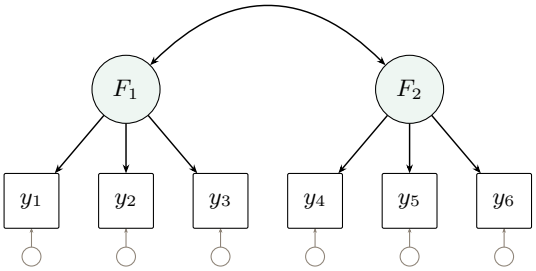
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Fix $\psi_{12} = 0$ in B (Lemma R).

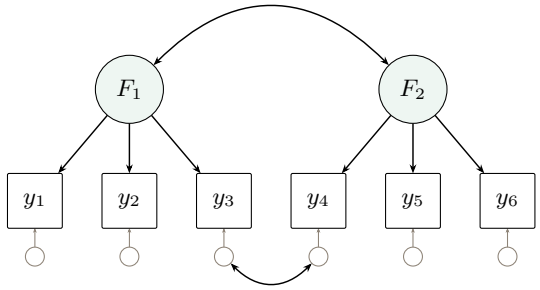
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 13 at the rational point listed in Appendix B, while model A has only $q_A = 12$ free parameters. □

In the game: Orthogonal factors are correlated factors with the correlation fixed to zero: A is nested in B.

E11 One residual covariance more

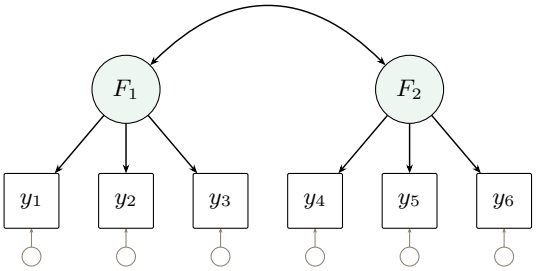
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y3 ~~ y4
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 14 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

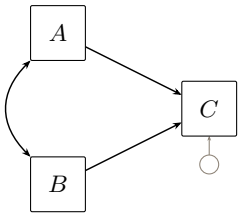
(ii) $B \preceq A$. Fix $\theta_{34} = 0$ in A (Lemma R). □

In the game: A is B plus one residual covariance (y3 ↔ y4). Fix it to zero and you get B, so B is nested in A.

E12 Correlated or independent predictors

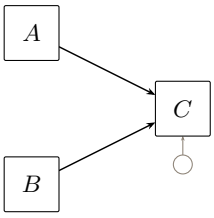
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 6$




```
C ~ A + B; A ~~ B
```

Model B $q_B = 5$




```
C ~ A + B
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 6 at the rational point listed in Appendix B, while model B has only $q_B = 5$ free parameters.

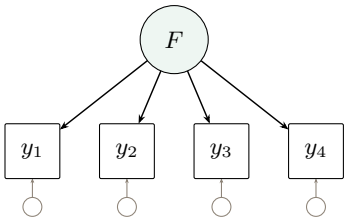
(ii) $B \preceq A$. Fix $\mathrm{Cov}(A, B) = 0$ in A (Lemma R). □

In the game: B is A with the covariance between the predictors fixed to zero (A and B independent). B is nested in A.

E13 Equal loadings

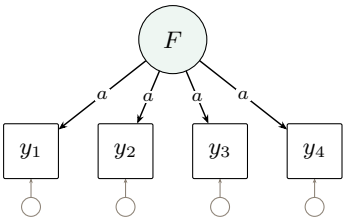
Level: Easy. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
F =~ y1 + y2 + y3 + y4
```

Model B $q_B = 6$




```
F =~ a*y1 + a*y2 + a*y3 + a*y4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 8 at the rational point listed in Appendix B, while model B has only $q_B = 6$ free parameters.

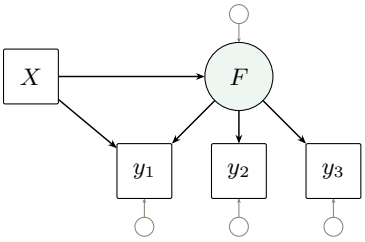
(ii) $B \preceq A$. B is A with the equality constraints $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4$ (Lemma R; with free factor variance the common loading can be set to 1). □

In the game: B forces all loadings to be equal (label a). That is a special case of A, where the loadings are free: B is nested in A.

E14 A direct effect of the covariate

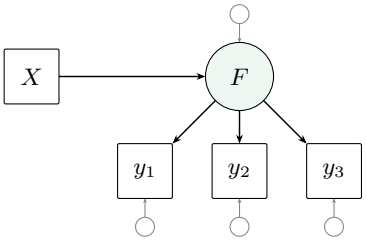
Level: Easy. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3; F ~ X; y1 ~ X
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3; F ~ X
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 9 at the rational point listed in Appendix B, while model B has only $q_B = 8$ free parameters.

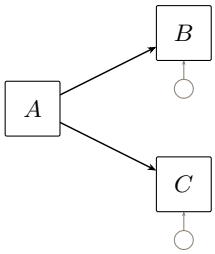
(ii) $B \preceq A$. Fix the direct effect $X \to y_1$ in A to 0 (Lemma R). □

In the game: A adds a direct effect of the covariate X on item y1. Fixing it to zero gives B, so B is nested in A.

E15 The independence model

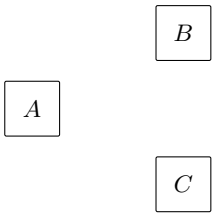
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ A
```

Model B $q_B = 3$




```
(no paths:  variances only)
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 5 at the rational point listed in Appendix B, while model B has only $q_B = 3$ free parameters.

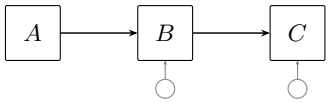
(ii) $B \preceq A$. Fix both regression coefficients in A to 0 (Lemma R): B only has variances. □

In the game: B has no paths at all: only variances (the independence or "baseline" model). It is nested in every model with the same variables, including A.

E16 Chain versus collider

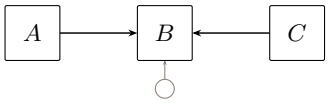
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 5$




```
B ~ A + C
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{AC}$. In B, A and C are exogenous and not connected, so $\sigma_{AC} = 0$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \beta_{BA}\beta_{CB}\,\mathrm{Var}(A) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3000 \neq 0$.

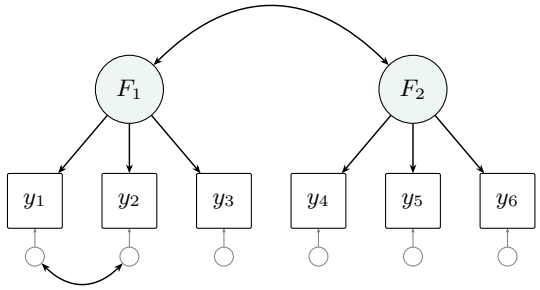
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{AC}\sigma_{BB} - \sigma_{AB}\sigma_{BC}$. In A, $\sigma_{AC} = \beta_{CB}\sigma_{AB}$ and $\sigma_{BC} = \beta_{CB}\sigma_{BB}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = -\sigma_{AB}\sigma_{BC} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = -0.2700 \neq 0$. □

In the game: The chain says A and C are independent given B; the collider says A and C are independent (period), but dependent given B. Each restriction is violated by the other model.

E17 Two different residual covariances

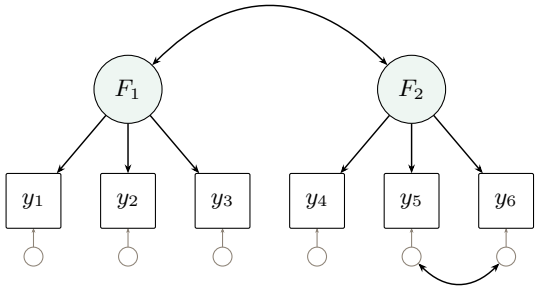
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y1 ~~ y2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y5 ~~ y6
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{35} - \sigma_{13}\sigma_{25}$. In B these four covariances are products $\lambda_i\lambda_j\psi_{f(i)f(j)}$ (no residual covariance involved), and the two products are equal. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{12}\lambda_3\lambda_5\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.06480 \neq 0$.

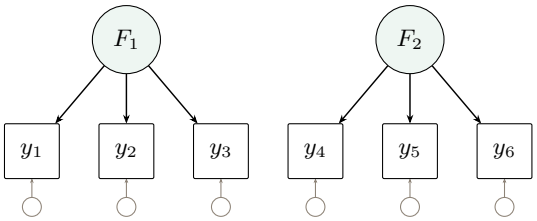
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{56}\sigma_{24} - \sigma_{45}\sigma_{26}$. In A these covariances are products of loadings and factor covariances (no θ_{56}). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{56}\lambda_2\lambda_4\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.06000 \neq 0$. □

In the game: Same number of parameters, but each model frees a residual covariance that the other fixes to zero. Neither can mimic the other.

E18 Two orthogonal factors versus one factor

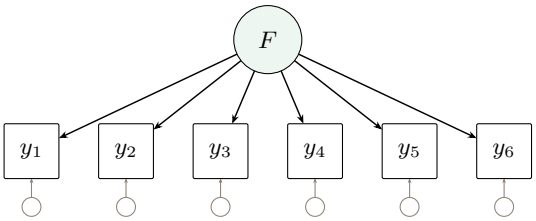
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6
```

Model B $q_B = 12$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{45} - \sigma_{14}\sigma_{25}$. One-factor model: every off-diagonal $\sigma_{ij} = \lambda_i \lambda_j \psi$, so all tetrads vanish. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_2\lambda_4\lambda_5\psi_{11}\psi_{22} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.440 \neq 0$.

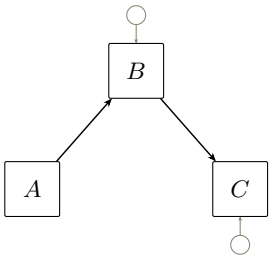
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{14}$. Orthogonal factors: the two clusters are uncorrelated. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \lambda_1\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.200 \neq 0$. □

In the game: A forces the two clusters to be uncorrelated; B forces all items through one factor, so the clusters must correlate. Neither restriction holds in the other model.

E19 Two different mediators

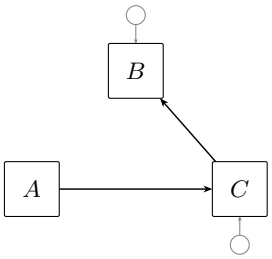
Level: Easy. Observed variables: A, B, C ($p = 3$, 6 moments).

Model A $q_A = 5$




```
B ~ A; C ~ B
```

Model B $q_B = 5$




```
C ~ A; B ~ C
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{AB}\sigma_{CC} - \sigma_{AC}\sigma_{BC}$. In B ($A \to C \to B$), A and B are independent given C. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.4500 \neq 0$.

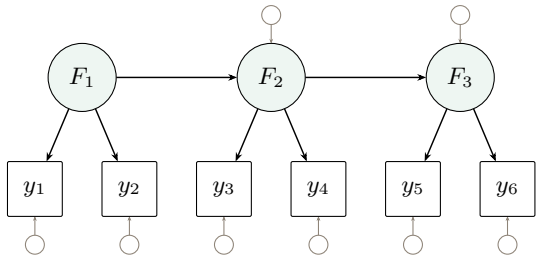
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{AC}\sigma_{BB} - \sigma_{AB}\sigma_{BC}$. In A ($A \to B \to C$), A and C are independent given B. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.6000 \neq 0$. □

In the game: Different mediators: A says A and C are independent given B, B says A and B are independent given C. Both are chains, but they make different claims.

E20 Latent chain versus latent collider

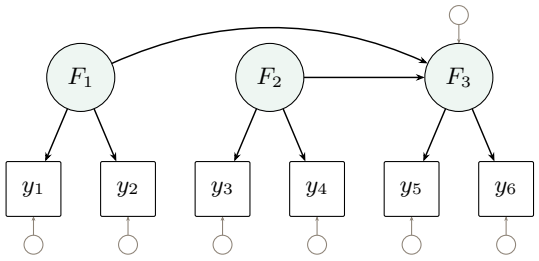
Level: Easy. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F3 ~ F1 + F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}$. In B, F_1 and F_2 are exogenous and uncorrelated, so indicators of F_1 and F_2 are uncorrelated. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_3\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.5000 \neq 0$.

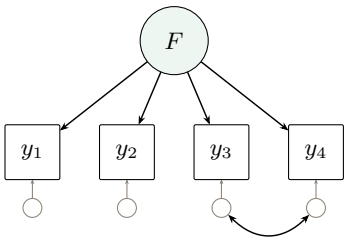
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{15}\sigma_{34} - \sigma_{13}\sigma_{45}$. In A this equals $\lambda_1\lambda_3\lambda_4\lambda_5(\psi_{13}\psi_{22} - \psi_{12}\psi_{23})$, and the chain implies $\psi_{13}\psi_{22} = \psi_{12}\psi_{23}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.5400 \neq 0$.□

In the game: A chain (F1 and F3 independent given F2) against a collider (F1 and F2 independent). Each model has a restriction the other violates.

M01 Residual covariance versus a second factor

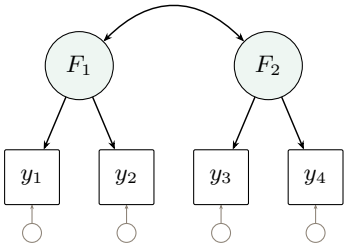
Level: Medium. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3 + y4; y3 ~~ y4
```

Model B $q_B = 9$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Lemma P with the pair (y_3, y_4): $A \equiv F\{y_1, y_2\} \to G\{y_3, y_4\}$. The latent block (F, G) is saturated in both that model and B (Lemma B(b), as in E03), so $A \equiv B$. Both equal $\{\Sigma \succ 0 : \sigma_{13}\sigma_{24} = \sigma_{14}\sigma_{23}\}$.

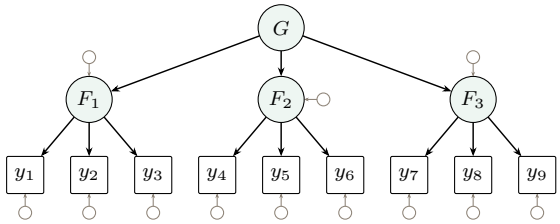
(ii) $B \preceq A$. See above. □

In the game: Both models imply a single restriction, the tetrad $\sigma 13{\cdot}\sigma 24 = \sigma 14{\cdot}\sigma 23$, and nothing else (9 parameters each). The residual covariance does the same job as splitting off a second factor.

M02 Higher-order model with three first-order factors

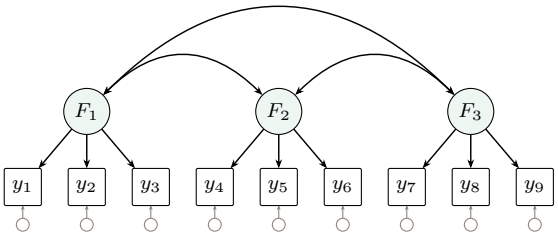
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; F1
~~ F2 + F3; F2 ~~ F3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part. In B, Ψ is free. In A, $\Psi = \gamma\gamma^\top \psi_G + \mathrm{diag}(\zeta)$ is a one-factor model on three variables, which by Lemma B(a) reproduces every Ψ with $\psi_{12}\psi_{13}\psi_{23} \neq 0$. Lemmas B(b) and G give $A \equiv B$.

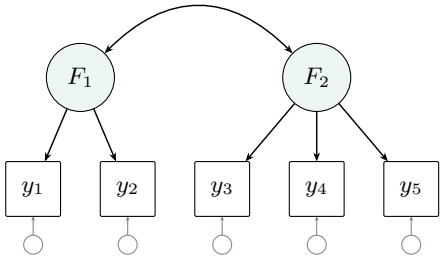
(ii) $B \preceq A$. See above (with $\gamma_1 = 1$, $\psi_G = \psi_{12}\psi_{13}/\psi_{23}$, ...). □

In the game: A second-order factor on three first-order factors has three loadings for three factor covariances: it is just-identified, so it reproduces any correlation pattern among three factors.

M03 A two-indicator factor

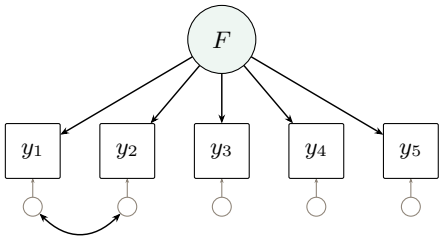
Level: Medium. Observed variables: y_1, y_2, y_3, y_4, y_5 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4 + y5; F1 ~~ F2
```

Model B $q_B = 11$




```
F =~ y1 + y2 + y3 + y4 + y5; y1 ~~ y2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. A has a factor F_1 with only y_1, y_2 and a free covariance with F_2. By Lemma P (pair (y_1, y_2)), $B \equiv F\{y_3, y_4, y_5\} \to G\{y_1, y_2\}$, and $F \to G$ and $F \leftrightarrow G$ give the same (saturated) 2×2 latent block (Lemma B(b)).

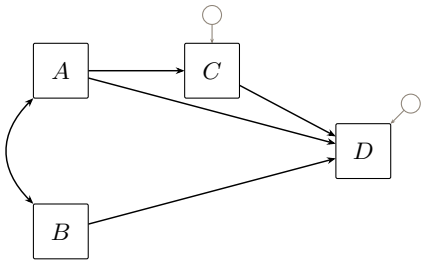
(ii) $B \preceq A$. See above. □

In the game: A factor with only two indicators adds exactly one free covariance (between those two items) on top of a single common factor. That is what the residual covariance y1 ↔ y2 does in B.

M04 The replacing rule

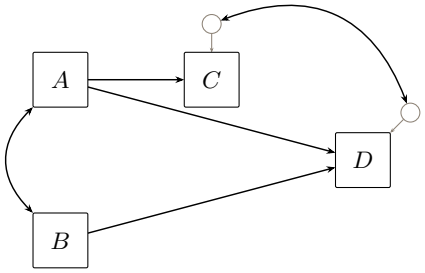
Level: Medium. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 9$




```
A ~~ B; C ~ A; D ~ C + A + B
```

Model B $q_B = 9$




```
A ~~ B; C ~ A; D ~ A + B; C ~~ D
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Substitute $C = \beta_{CA}A + \zeta_C$ in D's equation of A: $D = (\beta_{DA} + \beta_{DC}\beta_{CA})A + \beta_{DB}B + (\beta_{DC}\zeta_C + \zeta_D)$. This is model B with residual $\zeta'_D = \beta_{DC}\zeta_C + \zeta_D$, $\mathrm{Var}(\zeta'_D) = \beta^2_{DC}\psi_C + \psi_D$ and $\mathrm{Cov}(\zeta_C, \zeta'_D) = \beta_{DC}\psi_C$.

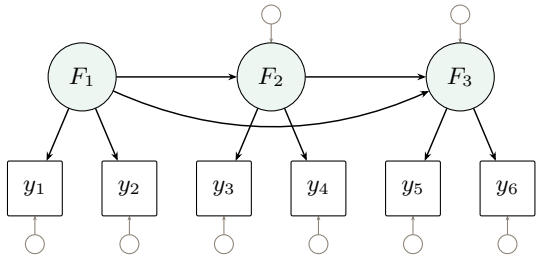
(ii) $B \preceq A$. Conversely, for $\psi_C \neq 0$ set $\beta_{DC} = \omega/\psi_C$ ($\omega = \mathrm{Cov}(\zeta_C, \zeta'_D)$), $\beta_{DA} = \beta'_{DA} - \beta_{DC}\beta_{CA}$, $\psi_D = \psi'_D - \omega^2/\psi_C$ (Lemma G). □

In the game: Replacing rule (Lee & Hershberger): C → D may be replaced by a residual covariance C ↔ D because the predictors of D include all predictors of C. Both models only claim that C and B are independent given A.

M05 A saturated structural block

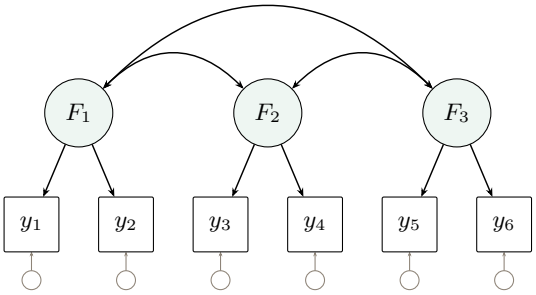
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F1 +
F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. The structural part of A is a complete recursive model on (F_1, F_2, F_3), which reproduces every $\Psi \succ 0$ (Cholesky, Lemma B(a)); in B, Ψ is free. Lemma B(b).

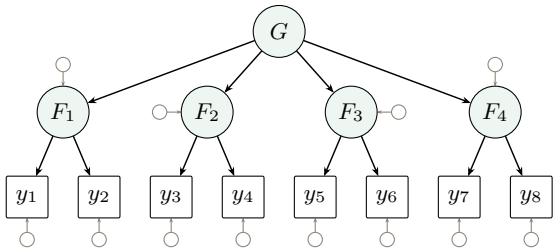
(ii) $B \preceq A$. See above. □

In the game: Three structural paths among three factors make a saturated latent block, just like three free factor covariances. Same measurement part, same fit.

M06 Higher-order model with four first-order factors

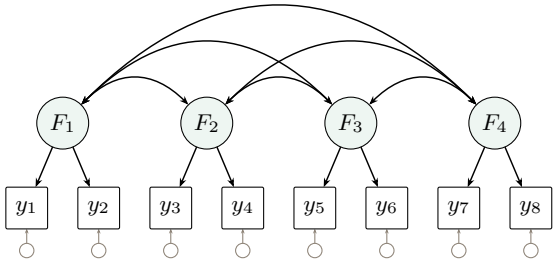
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Same measurement part; B has a free Ψ and A's second-order factor gives one particular Ψ (Lemma B(b)).

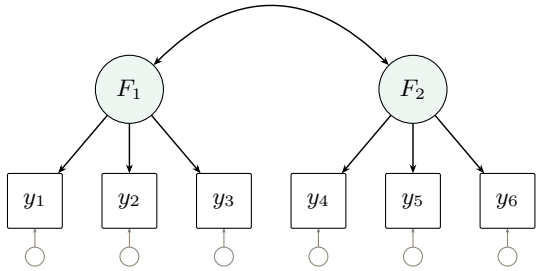
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 22 at the rational point listed in Appendix B, while model A has only $q_A = 20$ free parameters. □

In the game: With four first-order factors the second-order factor explains six covariances with four loadings, imposing two restrictions. B leaves the six covariances free: A is nested in B.

M07 Two factors inside one factor with correlated residuals

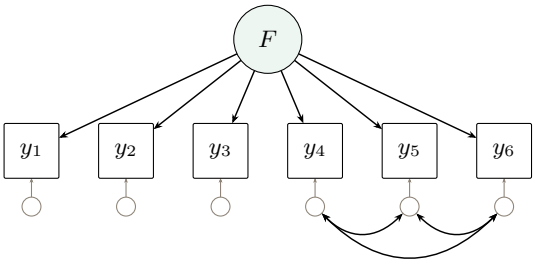
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y4 ~~ y5 + y6; y5 ~~ y6
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Given A ($\Psi = (\psi_{kl})$, $\psi_{11} \neq 0$), take in B: $\psi = \psi_{11}$, loadings λ_i for y_1, y_2, y_3 and $\lambda_j \psi_{12}/\psi_{11}$ for $j = 4, 5, 6$, $\theta_{jk} = \lambda_j \lambda_k (\psi_{22} - \psi_{12}^2/\psi_{11})$ for $j \neq k \in \{4, 5, 6\}$ and $\theta_{jj} = \theta_{jj}^A + \lambda_j^2 (\psi_{22} - \psi_{12}^2/\psi_{11})$. Then $\Sigma_B = \Sigma_A$ (Lemma G).

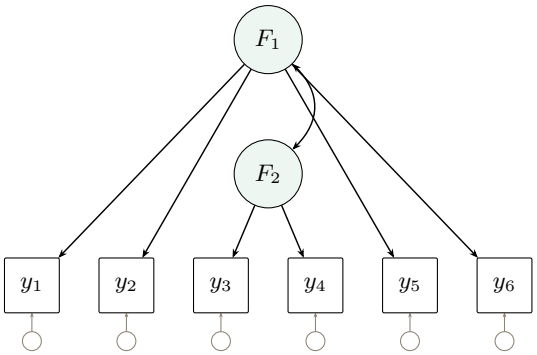
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 13$ free parameters. □

In the game: Surprise: one factor plus free residual covariances among y4–y6 can mimic the second factor. The part of y4–y6 that F1 does not explain goes into those residual covariances. B has two more parameters, so A is nested in B.

M08 Merging two factors

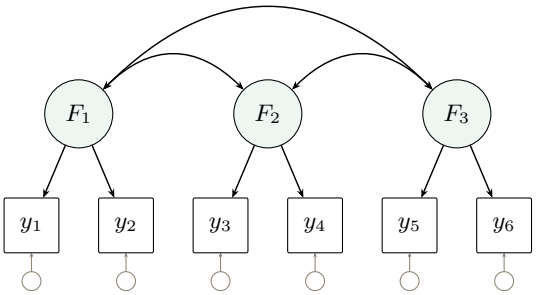
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y5 + y6; F2 =~ y3 + y4; F1 ~~ F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma M: merge F_1 and F_3 of B ($\psi_{11} = \psi_{33} = \psi_{13}$, $\psi_{12} = \psi_{23}$).

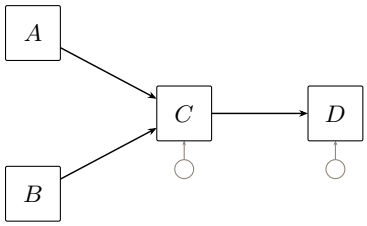
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 13$ free parameters. □

In the game: Merge the first and third factor of B (correlation 1) and you get A. So A, with its non-adjacent indicators, is nested in the three-factor model.

M09 Every model is nested in a saturated model

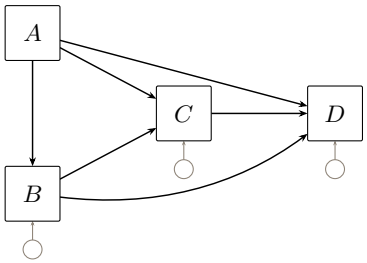
Level: Medium. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
C ~ A + B; D ~ C
```

Model B $q_B = 10$




```
B ~ A; C ~ A + B; D ~ A + B + C
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. B is a complete recursive path model on four variables, hence saturated (Lemma B(a)).

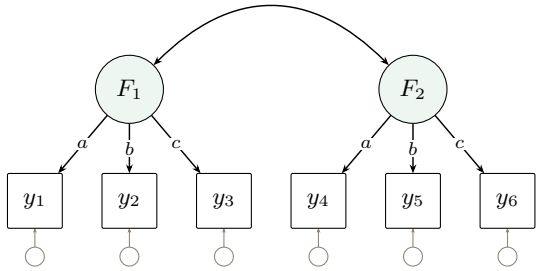
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 10 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: B has every possible path between the four variables: it is saturated. Any model on the same variables is nested in a saturated model.

M10 Equal loading patterns

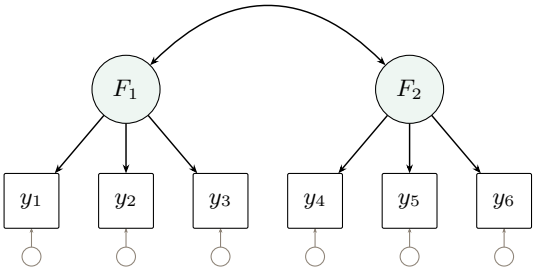
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ a*y1 + b*y2 + c*y3; F2 =~ a*y4 + b*y5 + c*y6; F1 ~~ F2
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. A is B with the constraints $\lambda_1 = \lambda_4$, $\lambda_2 = \lambda_5$, $\lambda_3 = \lambda_6$ (Lemma R).

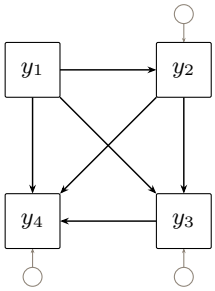
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 13 at the rational point listed in Appendix B, while model A has only $q_A = 12$ free parameters. □

In the game: A forces the same loading pattern (a, b, c) on both factors; B leaves all loadings free. A is a constrained version of B.

M11 A factor model inside a saturated path model

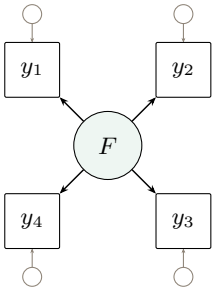
Level: Medium. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 10$




```
y2 ~ y1; y3 ~ y1 + y2; y4 ~ y1 + y2 + y3
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 10 at the rational point listed in Appendix B, while model B has only $q_B = 8$ free parameters.

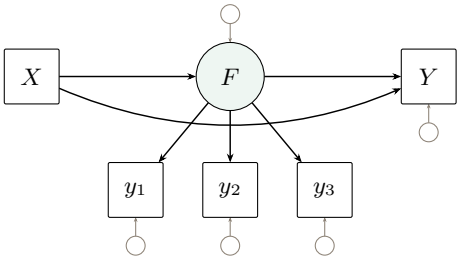
(ii) $B \preceq A$. A is a complete recursive path model on $y_1, \ldots, y_4$: saturated (Lemma B(a)). □

In the game: A is a saturated path model (all six paths). Every model on these four variables, also a factor model, is nested in it.

M12 Full versus partial mediation

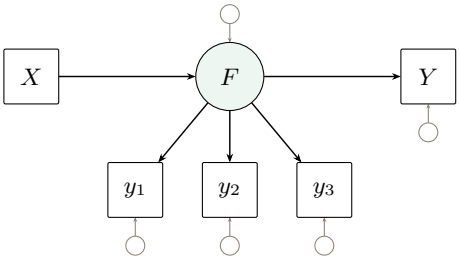
Level: Medium. Observed variables: X, y_1, y_2, y_3, Y ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F =~ y1 + y2 + y3; F ~ X; Y ~ F + X
```

Model B $q_B = 10$




```
F =~ y1 + y2 + y3; F ~ X; Y ~ F
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 11 at the rational point listed in Appendix B, while model B has only $q_B = 10$ free parameters.

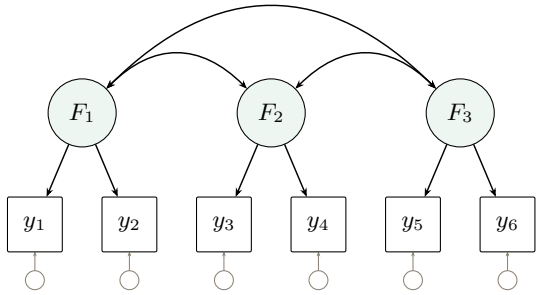
(ii) $B \preceq A$. Fix the direct effect $X \to Y$ in A to 0 (Lemma R). □

In the game: Partial mediation (A) against full mediation (B). Full mediation is partial mediation with the direct effect X → Y fixed to zero.

M13 A residual covariance as a factor

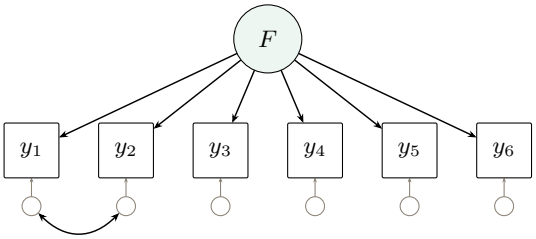
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Model B $q_B = 13$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 15 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

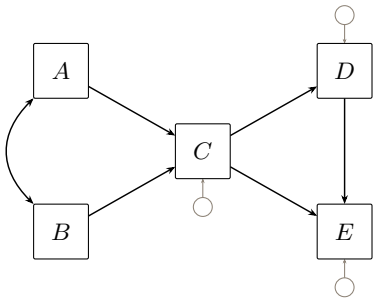
(ii) $B \preceq A$. By Lemma P, $B \equiv F\{y_3, \ldots, y_6\} \to G\{y_1, y_2\}$, which by Lemma B(b) is the two-factor model $\{y_1, y_2\} \,|\, \{y_3, \ldots, y_6\}$ with correlated factors. Merging F_2 and F_3 of A gives that model (Lemma M). Hence $B \preceq A$. □

In the game: A residual covariance between y1 and y2 acts like a separate two-indicator factor for them. So B is a two-factor model, which in turn is the three-factor model with F2 and F3 merged.

M14 Dropping one path

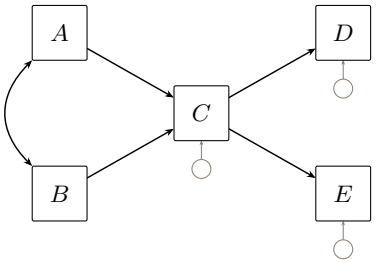
Level: Medium. Observed variables: A, B, C, D, E ($p = 5$, 15 moments).

Model A $q_A = 11$




```
A ~~ B; C ~ A + B; D ~ C; E ~ C + D
```

Model B $q_B = 10$




```
A ~~ B; C ~ A + B; D ~ C; E ~ C
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 11 at the rational point listed in Appendix B, while model B has only $q_B = 10$ free parameters.

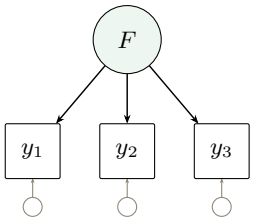
(ii) $B \preceq A$. Fix $\beta_{ED} = 0$ in A (Lemma R). □

In the game: B is A without the path D → E. Fix that coefficient to zero and A becomes B.

M15 A chain inside a saturated factor model

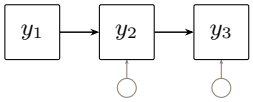
Level: Medium. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 6$




```
F =~ y1 + y2 + y3
```

Model B $q_B = 5$




```
y2 ~ y1; y3 ~ y2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 6 at the rational point listed in Appendix B, while model B has only $q_B = 5$ free parameters.

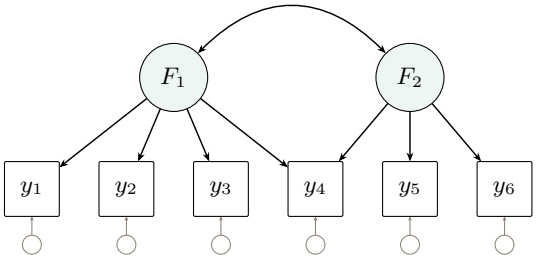
(ii) $B \preceq A$. The one-factor model with three indicators is saturated (Lemma B(a), see E05). □

In the game: The one-factor model with three indicators is saturated, so it contains every model for these three variables, including the chain y1 → y2 → y3.

M16 Cross-loadings on different items

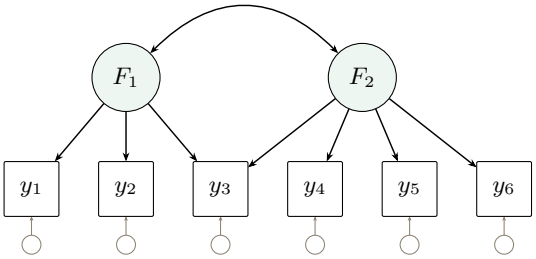
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y3 + y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{56}\sigma_{24} - \sigma_{46}\sigma_{25}$. In B, y_4, y_5, y_6 load only on F_2 and y_2 only on F_1, so the four covariances are $\lambda\lambda\psi$ products that cancel. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_2\lambda_4\lambda_5\lambda_6(\psi_{11}\psi_{22} - \psi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.172 \neq 0$.

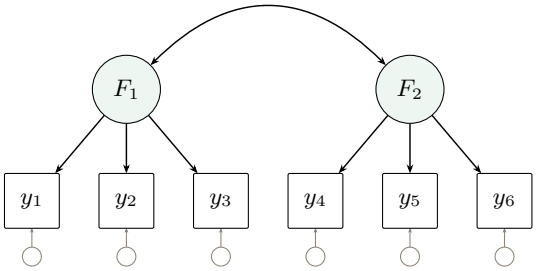
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{35} - \sigma_{13}\sigma_{25}$. In A, y_1, y_2, y_3 load only on F_1 and y_5 only on F_2. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.8880 \neq 0$. □

In the game: The cross-loading sits on a different item (y4 in A, y3 in B). Each model has a loading that the other fixes to zero.

M17 Different splits of the items

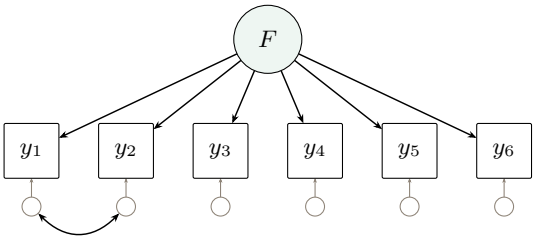
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 13$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{45} - \sigma_{14}\sigma_{35}$. In B these covariances do not involve θ_{12}, so they are $\lambda_i\lambda_j\psi$ and the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_3\lambda_4\lambda_5(\psi_{11}\psi_{22} - \psi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.199 \neq 0$.

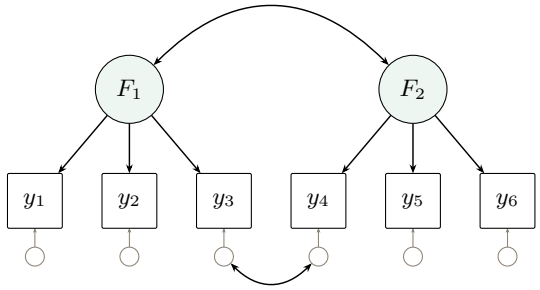
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{14}\sigma_{23}$. In A: $\lambda_1\lambda_2\psi_{11} \cdot \lambda_3\lambda_4\psi_{12} - \lambda_1\lambda_4\psi_{12} \cdot \lambda_2\lambda_3\psi_{11} = 0$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{12}\lambda_3\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3240 \neq 0$. □

In the game: B behaves like a two-factor model with y1, y2 as a separate factor; A splits the items as y1, y2, y3 and y4, y5, y6. Different splits: neither is nested.

M18 Residual covariance versus cross-loading

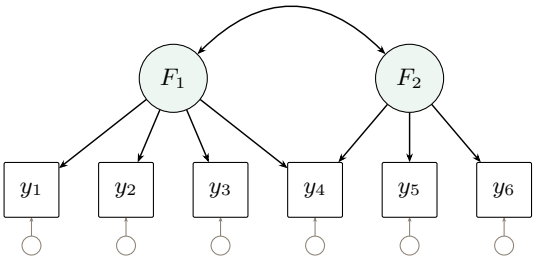
Level: Medium. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2; y3 ~~ y4
```

Model B $q_B = 14$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{34}\sigma_{15} - \sigma_{14}\sigma_{35}$. In B, y_4's covariances with y_1 and y_3 are both $\lambda_i(\lambda_4\psi_{12} + \kappa\psi_{11})$, so the rows of y_1 and y_3 stay proportional. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{34}\lambda_1\lambda_5\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.07200 \neq 0$.

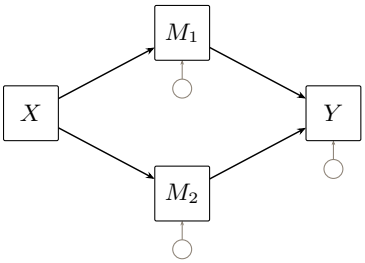
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{14}\sigma_{56} - \sigma_{15}\sigma_{46}$. In A, y_4 loads only on F_2 (the residual covariance only touches σ_{34}). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \lambda_1\lambda_5\lambda_6\kappa(\psi_{11}\psi_{22} - \psi_{12}^2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 1.172 \neq 0$. □

In the game: A residual covariance only adds covariance between y3 and y4. A cross-loading changes the covariances of y4 with all items. Same number of parameters, different restrictions.

M19 Parallel versus serial mediation

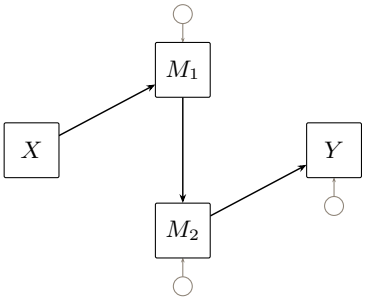
Level: Medium. Observed variables: X, M_1, M_2, Y ($p = 4$, 10 moments).

Model A $q_A = 8$




```
M1 ~ X; M2 ~ X; Y ~ M1 + M2
```

Model B $q_B = 7$




```
M1 ~ X; M2 ~ M1; Y ~ M2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 8 at the rational point listed in Appendix B, while model B has only $q_B = 7$ free parameters.

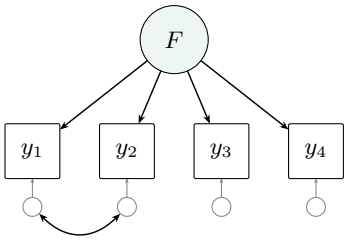
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{XX} - \sigma_{X1}\sigma_{X2}$. In A, M_1 and M_2 are independent given X (their only common cause). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.7200 \neq 0$. □

In the game: Parallel mediators (A: M1 and M2 independent given X) against serial mediation (B: M1 → M2). A has more parameters but still cannot produce B.

M20 Residual covariance on different pairs

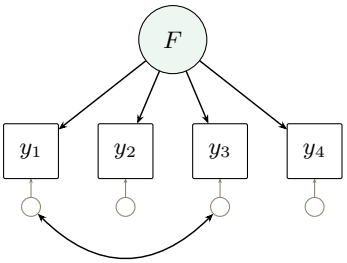
Level: Medium. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y2
```

Model B $q_B = 9$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{14}\sigma_{23}$. In B only σ_{13} contains a residual covariance; this tetrad avoids it. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{12}\lambda_3\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3240 \neq 0$.

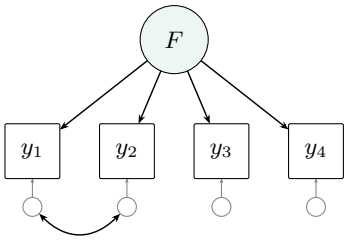
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{24} - \sigma_{14}\sigma_{23}$. In A only σ_{12} contains a residual covariance; this tetrad avoids it. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \theta_{13}\lambda_2\lambda_4\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3600 \neq 0$. □

In the game: Each model keeps a different tetrad: A implies $\sigma 13{\cdot}\sigma 24 = \sigma 14{\cdot}\sigma 23$, B implies $\sigma 12{\cdot}\sigma 34 = \sigma 14{\cdot}\sigma 23$. Neither model satisfies the other's restriction.

H01 Equivalent residual covariances

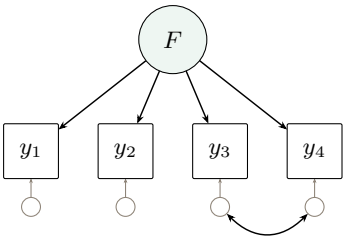
Level: Hard. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y2
```

Model B $q_B = 9$




```
F =~ y1 + y2 + y3 + y4; y3 ~~ y4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. By Lemma P, $A \equiv F\{y_3, y_4\} \to G\{y_1, y_2\}$ and $B \equiv F\{y_1, y_2\} \to G\{y_3, y_4\}$; by Lemma B(b) both are the two-factor model $\{y_1, y_2\}|\{y_3, y_4\}$ with correlated factors. Hence $A \equiv B$; both equal $\{\Sigma \succ 0 : \sigma_{13}\sigma_{24} = \sigma_{14}\sigma_{23}\}$.

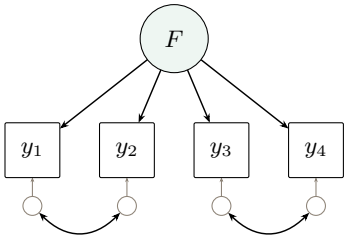
(ii) $B \preceq A$. See above. □

In the game: Both are the same as a two-factor model y1, y2 | y3, y4: either residual covariance frees the covariance within one pair, and the factor variance takes care of the other pair. Both imply only $\sigma 13 \cdot \sigma 24 = \sigma 14 \cdot \sigma 23$.

H02 A counting trap

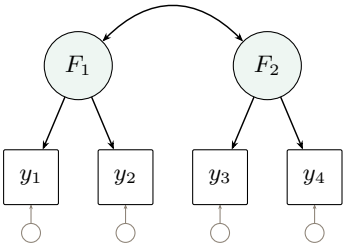
Level: Hard. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 10$




```
F =~ y1 + y2 + y3 + y4; y1 ~~ y2; y3 ~~ y4
```

Model B $q_B = 9$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Apply Lemma P to both pairs: $A \equiv G_1\{y_1, y_2\} \leftarrow F \rightarrow G_2\{y_3, y_4\}$ with F without indicators. The observed covariances depend on the latent block (G_1, G_2) only, which is saturated ($\mathrm{Cov}(G_1, G_2) = \gamma_1\gamma_2\psi_F$, variances free). By Lemma B(b), $A \equiv B$. Note $q_A = 10$ but $\dim \mathcal{M}_A = 9$: A is not identified.

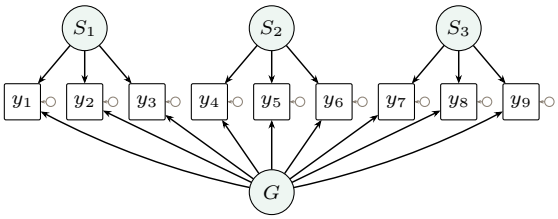
(ii) $B \preceq A$. See above. □

In the game: Counting trap: A has 10 parameters for 10 moments but is not identified, and still implies $\sigma 13 \cdot \sigma 24 = \sigma 14 \cdot \sigma 23$. It is exactly as restrictive as the two-factor model B.

H03 A three-indicator group factor equals a free residual block

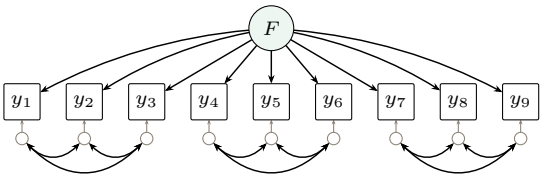
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 27$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; y1 ~~ y2 + y3;
y2 ~~ y3; y4 ~~ y5 + y6; y5 ~~ y6; y7 ~~ y8 + y9; y8 ~~ y9
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. In A, group factor S_k adds $s_i s_j \psi_{S_k}$ to σ_{ij} for $i \neq j$ in group k, and $s_i^2 \psi_{S_k}$ to σ_{ii}. Map to B: same general loadings, $\theta_{ij} = s_i s_j \psi_{S_k}$, $\theta_{ii} = \theta_{ii}^A + s_i^2 \psi_{S_k}$.

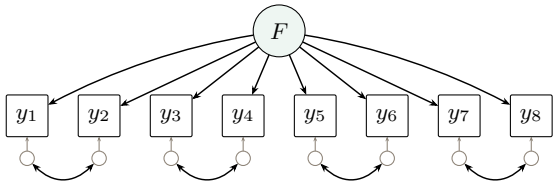
(ii) $B \preceq A$. Conversely, in group $k = \{a, b, c\}$ with $\theta_{ab}\theta_{ac}\theta_{bc} \neq 0$: $s_a = 1$, $\psi_{S_k} = \theta_{ab}\theta_{ac}/\theta_{bc}$, $s_b = \theta_{bc}/\theta_{ac}$, $s_c = \theta_{bc}/\theta_{ab}$ (Lemma B(a) on the residual block), residual variances adjusted. Lemma G. □

In the game: A group factor with three indicators has three loadings for the three covariances within its group: it can produce any pattern there. That is exactly what three free residual covariances do.

H04 One factor with paired residuals equals a higher-order model

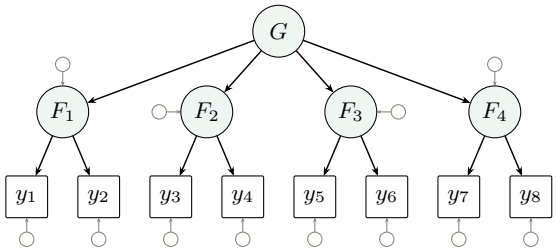
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y2; y3 ~~ y4;
y5 ~~ y6; y7 ~~ y8
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Lemma P applied to the four pairs: $A \equiv G_k\{y_{2k-1}, y_{2k}\} \leftarrow F$ ($k = 1, \ldots, 4$) with F without indicators. That is exactly the higher-order model B with second-order factor F.

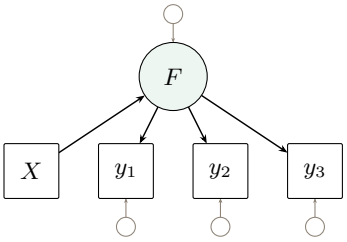
(ii) $B \preceq A$. See above. □

In the game: In both models all covariances between pairs follow one general factor, and the covariance within each pair is free (a residual covariance in A, a first-order factor variance in B). Same 20 free parameters, same restrictions.

H05 Covariate or indicator

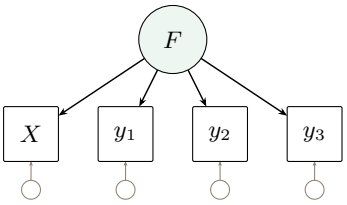
Level: Hard. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
F =~ y1 + y2 + y3; F ~ X
```

Model B $q_B = 8$




```
F =~ X + y1 + y2 + y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Given A ($F = \gamma X + \zeta$, $v = \mathrm{Var}(X)$): put $\psi = \gamma^2 v + \psi_\zeta$, keep λ_i, and $\lambda_X = \gamma v/\psi$, $\theta_X = v - \lambda_X^2 \psi$. Then $\sigma_{X,i} = \lambda_X \lambda_i \psi = \gamma v \lambda_i$ and $\sigma_{ij} = \lambda_i \lambda_j \psi$ as in A.

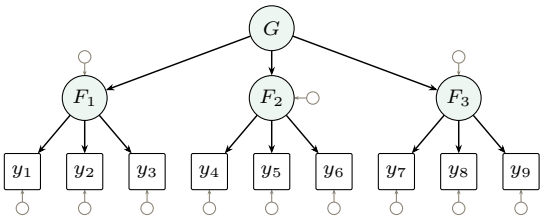
(ii) $B \preceq A$. Given B: $v = \sigma_{XX}$, $\gamma = \lambda_X \psi / v$, $\psi_\zeta = \psi - \gamma^2 v$ (Lemma G; $v \neq 0$). □

In the game: A covariate that predicts the factor (A) cannot be told apart from one more indicator of that factor (B). Both say that X relates to the items only through F.

H06 Higher-order inside bifactor

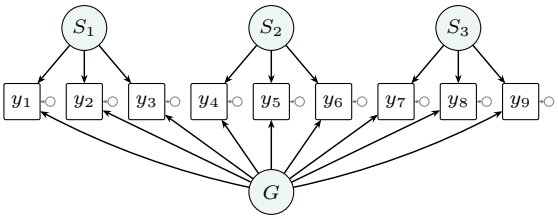
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma H (Schmid–Leiman; Yung, Thissen & McLeod, 1999).

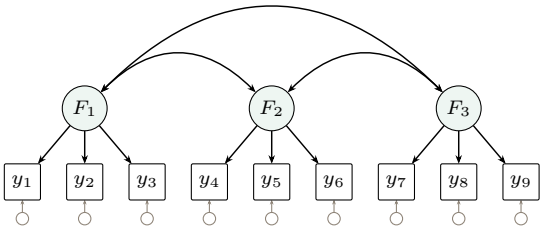
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 27 at the rational point listed in Appendix B, while model A has only $q_A = 21$ free parameters. □

In the game: Schmid–Leiman: a higher-order model is a bifactor model whose general and group loadings are proportional within each group (Yung, Thissen & McLeod, 1999). The bifactor model drops those proportionality constraints.

H07 Correlated factors inside bifactor

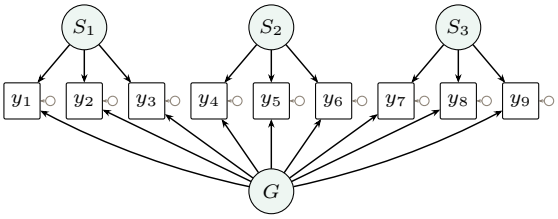
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; F1
~~ F2 + F3; F2 ~~ F3
```

Model B $q_B = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By M02, three correlated factors are equivalent to the higher-order model, which is nested in the bifactor model (Lemma H).

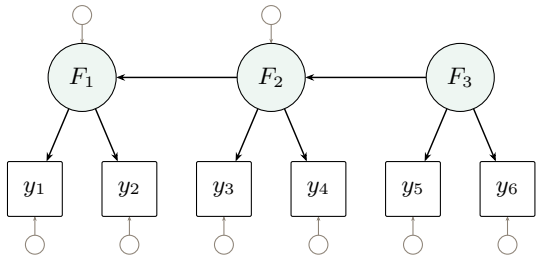
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 27 at the rational point listed in Appendix B, while model A has only $q_A = 21$ free parameters. □

In the game: Three correlated factors are equivalent to a higher-order model (just-identified), and a higher-order model is nested in the bifactor model. So the correlated-factors model is nested in the bifactor model.

H08 Arrows the other way, still nested

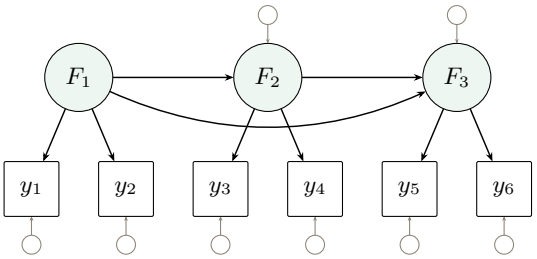
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F3; F1 ~ F2
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F1 +
F2
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. The structural part of B is a complete recursive model on (F_1, F_2, F_3), which reproduces every $\Psi \succ 0$ (Lemma B(a)), including the Ψ of A's chain $F_3 \to F_2 \to F_1$. Lemma B(b).

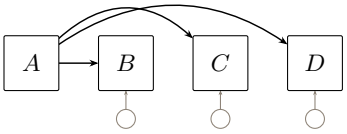
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: The arrows in B point the other way, but B's latent block is saturated: it reproduces any covariance matrix of the three factors, including the one implied by A's chain.

H09 A perfect indicator

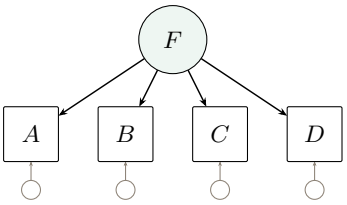
Level: Hard. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
B ~ A; C ~ A; D ~ A
```

Model B $q_B = 8$




```
F =~ A + B + C + D
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma T with A as the perfect indicator: set $\lambda_A = 1$, $\theta_A = 0$, $\psi = \mathrm{Var}(A)$ and $\lambda_k = \beta_{kA}$, $\theta_k = \psi_k$ ($k = B, C, D$).

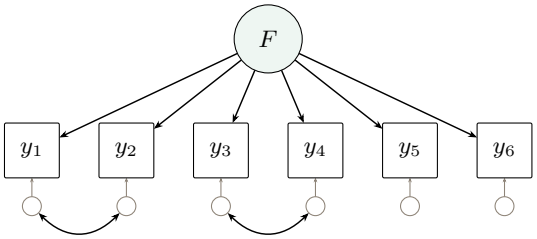
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: Let A be a perfect indicator of F (residual variance 0): then F is A, and the factor model becomes the path model with A as the common cause. A is nested in B.

H10 Paired residuals inside three factors

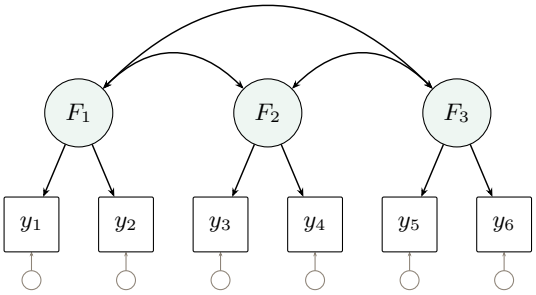
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2; y3 ~~ y4
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Lemma P twice: $A \equiv G_1\{y_1, y_2\} \leftarrow F\{y_5, y_6\} \rightarrow G_2\{y_3, y_4\}$, a three-factor model with a particular latent covariance matrix. In B, Ψ is free (Lemma B(b)).

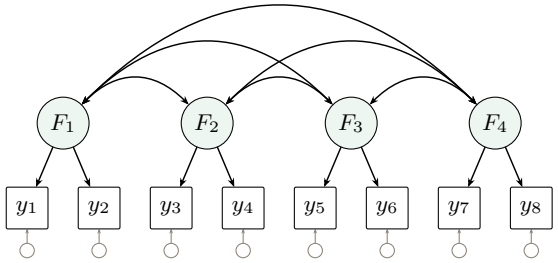
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: Each residual covariance within a pair acts like a two-indicator factor. A is a one-factor model with two such pairs; the three-factor model B also frees the third pair.

H11 Paired residuals inside four correlated factors

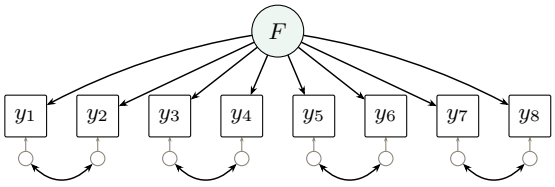
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Model B $q_B = 20$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8; y1 ~~ y2; y3 ~~ y4;
y5 ~~ y6; y7 ~~ y8
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 22 at the rational point listed in Appendix B, while model B has only $q_B = 20$ free parameters.

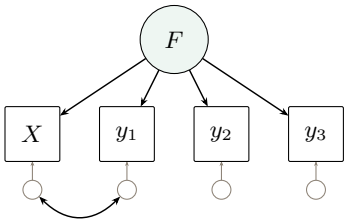
(ii) $B \preceq A$. By H04, B is equivalent to the higher-order model over the four pair factors, and that model is nested in A, where Ψ is free (Lemma B(b)). □

In the game: B is equivalent to a second-order model over four two-indicator factors (one general factor plus a free covariance within each pair), and that is nested in four freely correlated factors.

H12 A covariate inside a factor model with one residual covariance

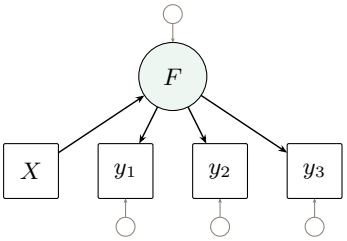
Level: Hard. Observed variables: X, y_1, y_2, y_3 ($p = 4$, 10 moments).

Model A $q_A = 9$




```
F =~ X + y1 + y2 + y3; X ~~ y1
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3; F ~ X
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 9 at the rational point listed in Appendix B, while model B has only $q_B = 8$ free parameters.

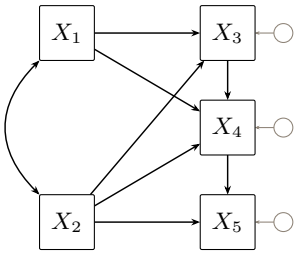
(ii) $B \preceq A$. By H05, B is equivalent to the one-factor model with indicators X, y_1, y_2, y_3, which is A with $\theta_{X1} = 0$ (Lemma R). □

In the game: B (X predicts F) is equivalent to the one-factor model with X as fourth indicator. A is that same model plus a residual covariance, so B is nested in A.

H13 Replacing rule plus one restriction

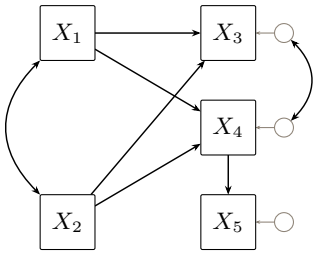
Level: Hard. Observed variables: X_1, X_2, X_3, X_4, X_5 ($p = 5$, 15 moments).

Model A $q_A = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X4 + X2; X1 ~~ X2
```

Model B $q_B = 12$




```
X3 ~ X1 + X2; X4 ~ X1 + X2; X5 ~ X4; X1 ~~ X2; X3 ~~ X4
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 13 at the rational point listed in Appendix B, while model B has only $q_B = 12$ free parameters.

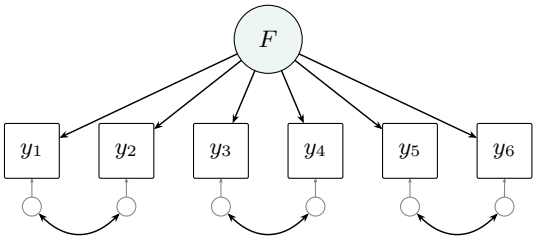
(ii) $B \preceq A$. As in M04 (with $C = X_3$, $D = X_4$; the equation of X_5 is unchanged), replacing $X_3 \to X_4$ by the residual covariance $X_3 \leftrightarrow X_4$ gives a model equivalent to A without $X_2 \to X_5$, which is nested in A (Lemma R). □

In the game: Swapping X3 → X4 for the residual covariance X3 ↔ X4 is allowed (same predictors), so that change alone gives an equivalent model. B also drops X2 → X5, which makes it nested in A.

H14 A two-indicator factor inside residual pairs

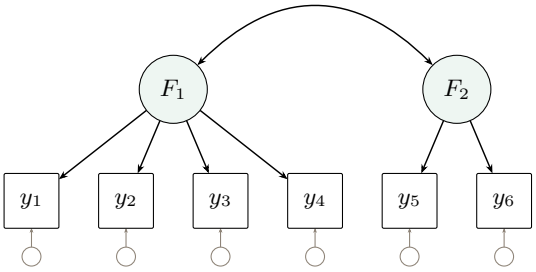
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2; y3 ~~ y4; y5 ~~ y6
```

Model B $q_B = 13$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y5 + y6; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 15 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

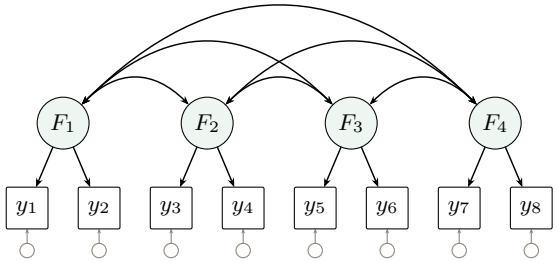
(ii) $B \preceq A$. By Lemma P and Lemma B(b), $B \equiv$ one factor plus θ_{56}. That is A with $\theta_{12} = \theta_{34} = 0$ (Lemma R). □

In the game: A two-indicator factor (F2 in B) is equivalent to one residual covariance on top of one factor. So B is one factor plus y5 ↔ y6, which is A with two of its residual covariances fixed to zero.

H15 Merging non-adjacent factors

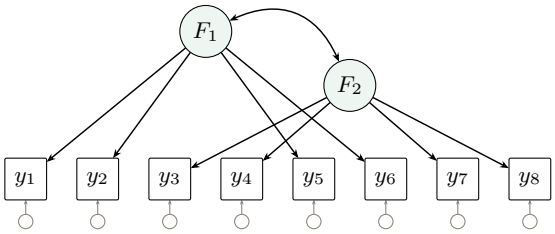
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 22$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F1
~~ F2 + F3 + F4; F2 ~~ F3 + F4; F3 ~~ F4
```

Model B $q_B = 17$




```
F1 =~ y1 + y2 + y5 + y6; F2 =~ y3 + y4 + y7 + y8; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 22 at the rational point listed in Appendix B, while model B has only $q_B = 17$ free parameters.

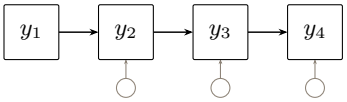
(ii) $B \preceq A$. Lemma M: in A merge F_1 with F_3 and F_2 with F_4. □

In the game: Merge F1 with F3 and F2 with F4 (correlations of 1) in the four-factor model and you get B. Non-adjacent indicators do not matter for nesting.

H16 Markov chain versus one factor

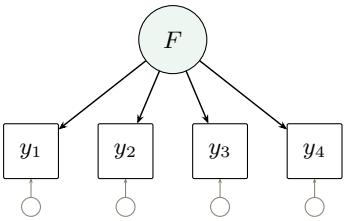
Level: Hard. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
y2 ~ y1; y3 ~ y2; y4 ~ y3
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{13}\sigma_{24}$. A tetrad, which vanishes in every one-factor model. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \sigma_{12}\sigma_{34}\big(1 - \sigma_{23}^2/(\sigma_{22}\sigma_{33})\big) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.1800 \neq 0$.

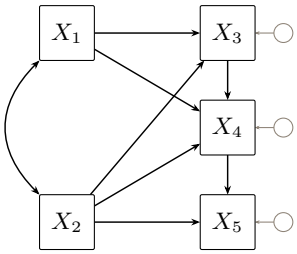
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: The chain makes y1 and y3 independent given y2; the factor model never makes items conditionally independent given another item. The factor model implies tetrads, which the chain violates.

H17 An illegal replacement

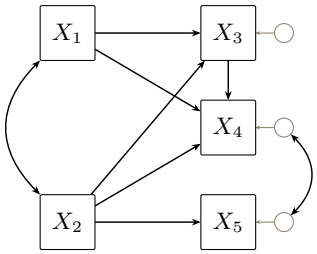
Level: Hard. Observed variables: X_1, X_2, X_3, X_4, X_5 ($p = 5$, 15 moments).

Model A $q_A = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X4 + X2; X1 ~~ X2
```

Model B $q_B = 13$




```
X3 ~ X1 + X2; X4 ~ X1 + X2 + X3; X5 ~ X2; X1 ~~ X2; X4 ~~ X5
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{51}\sigma_{22} - \sigma_{52}\sigma_{12}$. In B, $X_5 = \beta X_2 + \zeta_5$ and ζ_5 is correlated only with ζ_4, so X_5 and X_1 are independent given X_2. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.2747 \neq 0$.

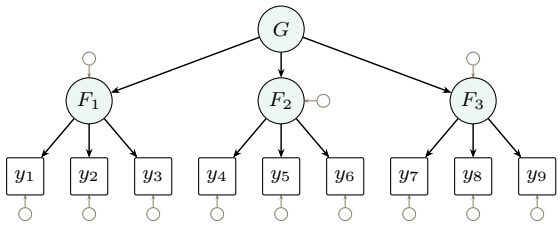
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{51} & \sigma_{52} & \sigma_{54} \\ \sigma_{21} & \sigma_{22} & \sigma_{24} \\ \sigma_{41} & \sigma_{42} & \sigma_{44} \end{pmatrix}$. In A, X_5 is independent of its non-descendant X_1 given its parents X_2, X_4 (local Markov property); this determinant is the corresponding partial covariance times $\det \Sigma_{\{2,4\}}$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.1221 \neq 0$. □

In the game: An illegal use of the replacing rule: X4 → X5 cannot become X4 ↔ X5, because X5's predictors (X2) do not include those of X4. The models make different independence claims about X5.

H18 Higher-order versus one factor with one residual block

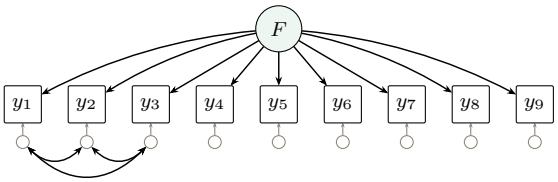
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9; G
=~ F1 + F2 + F3
```

Model B $q_B = 21$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9; y1 ~~ y2 + y3;
y2 ~~ y3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{45}\sigma_{17} - \sigma_{14}\sigma_{57}$. In B these four covariances do not involve residual covariances (those are only within y_1, y_2, y_3), so the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_4\lambda_5\lambda_7\gamma_1\gamma_3\psi_G\zeta_2 \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.9504 \neq 0$.

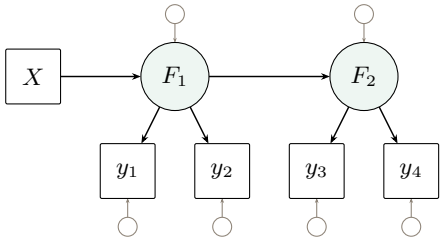
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{13}\sigma_{24}$. In A, y_1, y_2, y_3 load only on F_1 and y_4 only on F_2. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = \lambda_4\psi(\theta_{12}\lambda_3 - \theta_{13}\lambda_2) \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.08400 \neq 0$. □

In the game: B forces the items of the second and third cluster to follow one general factor exactly; A gives every cluster its own factor. But A restricts the first cluster, which B leaves free. Neither contains the other.

H19 Two mediation chains

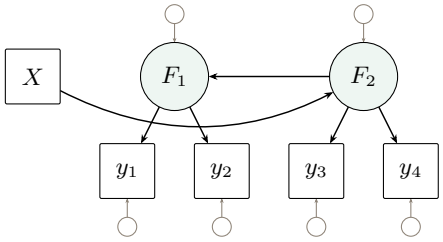
Level: Hard. Observed variables: X, y_1, y_2, y_3, y_4 ($p = 5$, 15 moments).

Model A $q_A = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~ X; F2 ~ F1
```

Model B $q_B = 11$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F2 ~ X; F1 ~ F2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{X1}\sigma_{34} - \sigma_{X3}\sigma_{14}$. In B ($X \to F_2 \to F_1$), X and F_1 are independent given F_2. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_3\lambda_4\,\gamma\,\mathrm{Var}(X)\,\zeta_2 \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.5400 \neq 0$.

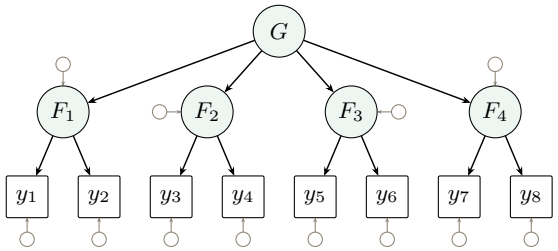
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{X3}\sigma_{12} - \sigma_{X1}\sigma_{23}$. In A ($X \to F_1 \to F_2$), X and F_2 are independent given F_1. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.5000 \neq 0$. □

In the game: Both are mediation chains starting at X, but with a different mediator. A says X and F2 are independent given F1; B says X and F1 are independent given F2.

H20 General factor versus latent chain

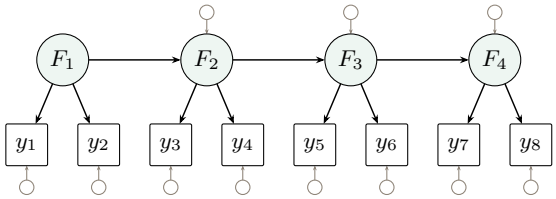
Level: Hard. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Model B $q_B = 19$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; F2
~ F1; F3 ~ F2; F4 ~ F3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 20 at the rational point listed in Appendix B, while model B has only $q_B = 19$ free parameters.

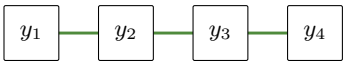
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{57} - \sigma_{15}\sigma_{37}$. This equals $\lambda\lambda\lambda\lambda(\psi_{12}\psi_{34} - \psi_{13}\psi_{24})$, which vanishes when Ψ follows one second-order factor. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = 0.1600 \neq 0$. □

In the game: Two ways to explain the correlations among four factors: one common cause (tetrad restrictions) or a chain (conditional independencies). Neither set of restrictions implies the other.

P01 A path network is a chain

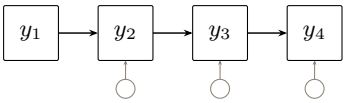
Level: Psychometrika. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4
```

Model B $q_B = 7$




```
y2 ~ y1; y3 ~ y2; y4 ~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Lemma C: the directed chain has skeleton $y_1 - y_2 - y_3 - y_4$ and no collider, so it equals the GGM on that path.

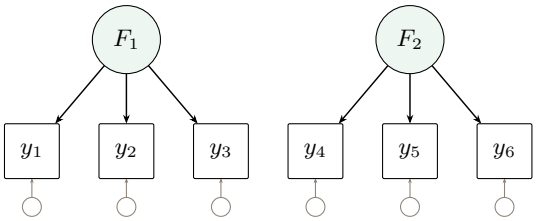
(ii) $B \preceq A$. See above. □

In the game: A Gaussian graphical model (GGM) whose graph is a path has the same conditional independencies as the directed chain: both are exactly the covariance matrices with a tridiagonal inverse.

P02 Orthogonal factors and network cliques

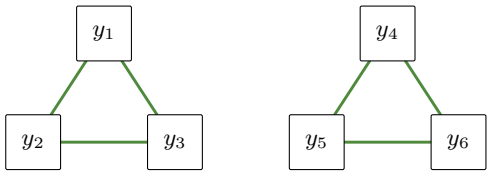
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 12$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6
```

Model B $q_B = 12$




```
network y1 y2 y3 y4 y5 y6; y1 - y2 + y3; y2 - y3; y4 - y5 + y6;
y5 - y6
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. $\mathcal{M}_A$: block-diagonal Σ with blocks $\{1,2,3\}, \{4,5,6\}$, each block a one-factor model with three indicators, which reproduces every positive definite block generically (Lemma B(a)). By Lemma I, Σ is block-diagonal iff Σ^{-1} is, which is $\mathcal{M}_B$ (two cliques, no edges between). Hence $\overline{\mathcal{M}_A} = \overline{\mathcal{M}_B}$.

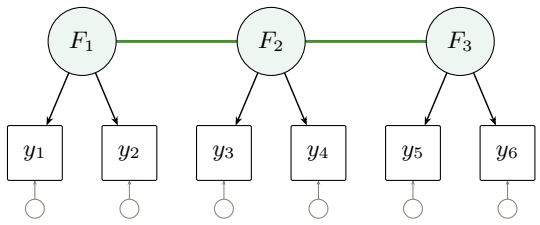
(ii) $B \preceq A$. See above. □

In the game: Orthogonal factors make the covariance matrix block-diagonal, and so is its inverse (two separate cliques in the GGM). With three indicators each factor is saturated within its block.

P03 Latent network chain equals latent chain

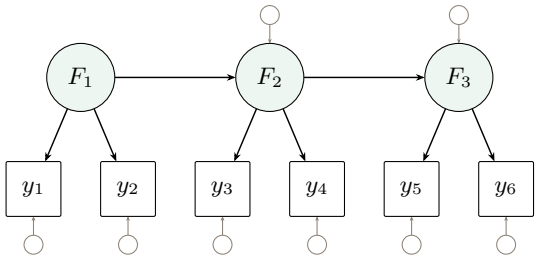
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; network F1 F2 F3;
F1 - F2; F2 - F3
```

Model B $q_B = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F2 ~ F1; F3 ~ F2
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same measurement part. In A, $\Psi = K^{-1}$ with $k_{13} = 0$, which by Lemma I means $\psi_{12}\psi_{23} = \psi_{13}\psi_{22}$; by Lemma C that is exactly the set of Ψ of the chain $F_1 \to F_2 \to F_3$. Lemma B(b).

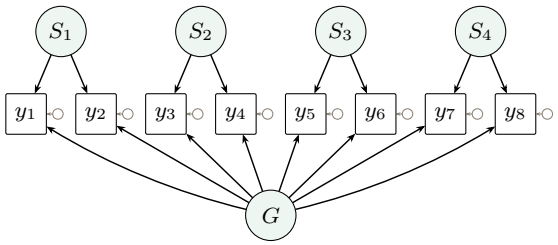
(ii) $B \preceq A$. See above. □

In the game: A latent network model (LNM) with edges F1 — F2 — F3 says F1 and F3 are independent given F2, exactly like the latent chain F1 → F2 → F3.

P04 Bifactor with two-indicator groups

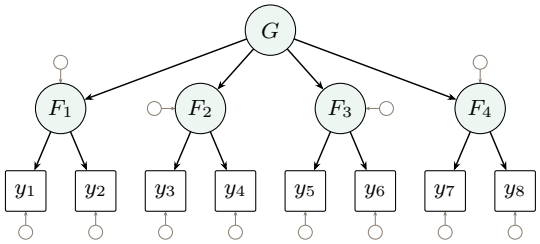
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 24$




```
S1 =~ y1 + y2; S2 =~ y3 + y4; S3 =~ y5 + y6; S4 =~ y7 + y8; G
=~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. A group factor with two indicators y_i, y_j only adds $s_i s_j \psi_S$ to σ_{ij} (and to the variances): it acts as a free residual covariance. So $A \equiv$ one factor (G) on $y_1, \ldots, y_8$ plus four within-pair residual covariances, which is equivalent to B by H04.

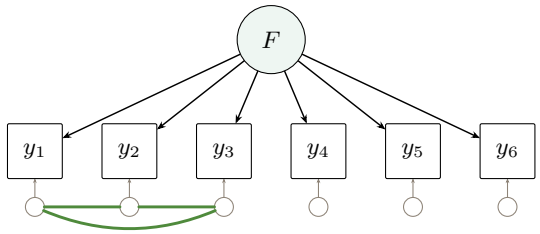
(ii) $B \preceq A$. See above. □

In the game: With two indicators per group factor, the bifactor model only frees the covariance within each pair on top of a general factor, and so does the higher-order model. Here the two models are equivalent, not merely nested!

P05 Residual clique equals residual covariance block

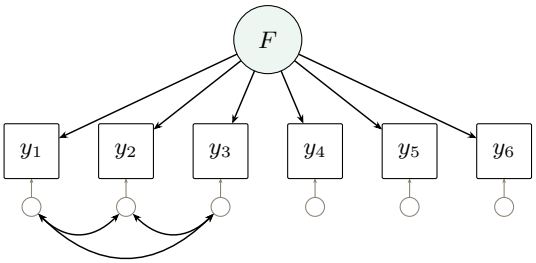
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3; y1 - y2 + y3; y2 - y3
```

Model B $q_B = 15$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2 + y3; y2 ~~ y3
```

Claim. $A \equiv B$: the models are **equivalent**.

Proof. (i) $A \preceq B$. Same factor part. The residual covariance matrix is $\Theta = K^{-1}$ in A and free on the block $\{1, 2, 3\}$ (diagonal elsewhere) in B; by Lemma I these are the same sets of Θ (block-diagonal inverse).

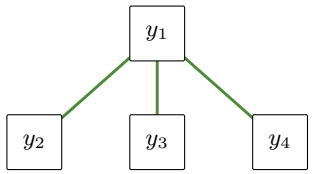
(ii) $B \preceq A$. See above. □

In the game: A residual network model (RNM): a fully connected block of residual partial correlations is the same as a fully free block of residual covariances (the inverse of a block-diagonal matrix is block-diagonal).

P06 Star network inside one factor

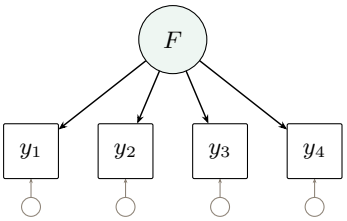
Level: Psychometrika. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 7$




```
network y1 y2 y3 y4; y1 - y2 + y3 + y4
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By Lemma C the star GGM equals the DAG $y_1 \to y_k$ ($k = 2, 3, 4$), which by Lemma T is the one-factor model with $\theta_1 = 0$.

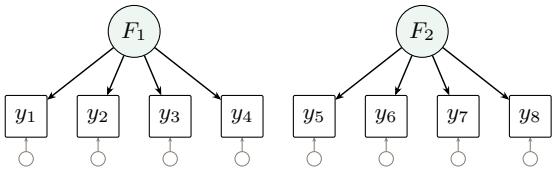
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: In the star network y1 explains all associations. Make y1 a perfect indicator of F (no residual variance) and the factor model reproduces the star: A is nested in B.

P07 Orthogonal factors inside network cliques

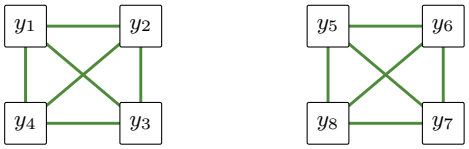
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 16$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y5 + y6 + y7 + y8
```

Model B $q_B = 20$




```
network y1 y2 y3 y4 y5 y6 y7 y8; y1 - y2 + y3 + y4; y2 - y3 +
y4; y3 - y4; y5 - y6 + y7 + y8; y6 - y7 + y8; y7 - y8
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. Orthogonal factors give a block-diagonal Σ, hence (Lemma I) a block-diagonal Σ^{-1}: no edges between the clusters, which is B.

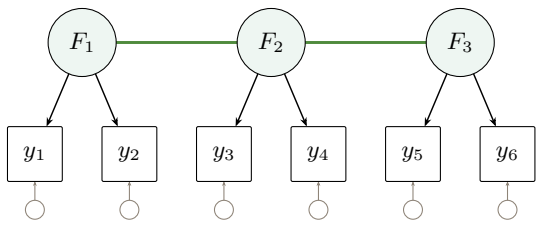
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 20 at the rational point listed in Appendix B, while model A has only $q_A = 16$ free parameters. □

In the game: Orthogonal factors give a block-diagonal covariance matrix, so its inverse is block-diagonal too: two fully connected clusters in the GGM. The GGM leaves each block free, the factors do not.

P08 Latent network chain inside a higher-order model

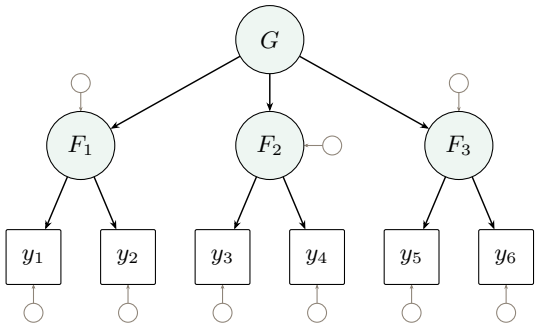
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; network F1 F2 F3;
F1 - F2; F2 - F3
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; G =~ F1 + F2 + F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By P03, A is the latent chain; its Ψ is a particular covariance matrix of (F_1, F_2, F_3). The second-order factor of B reproduces every Ψ with nonzero covariances (as in M02). Lemma B(b).

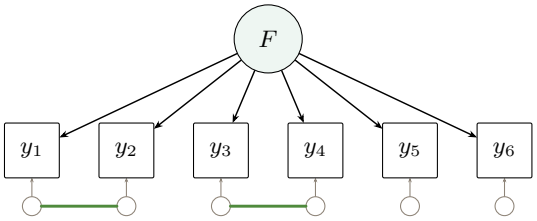
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: The latent chain network removes the F1 — F3 edge (one restriction). A second-order factor on three factors is just-identified, so it reproduces any latent covariance matrix.

P09 Isolated residual edges

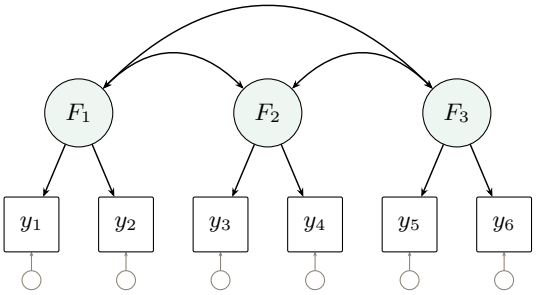
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3 y4; y1 - y2;
y3 - y4
```

Model B $q_B = 15$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F1 ~~ F2 + F3; F2
~~ F3
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. By Lemma I, a residual network with the two separate edges $y_1 - y_2$ and $y_3 - y_4$ has residual covariance matrix with free θ_{12}, θ_{34} and nothing else, so A is the model of H10, which is nested in B.

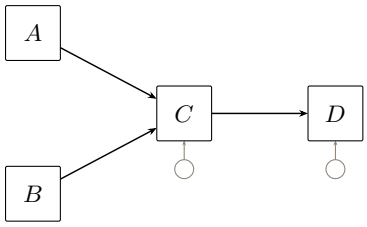
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 15 at the rational point listed in Appendix B, while model A has only $q_A = 14$ free parameters. □

In the game: Isolated residual edges in an RNM are just residual covariances. And a residual covariance within a pair acts like a two-indicator factor, so A is the three-factor model with one pair not separated.

P10 A DAG inside its moral graph

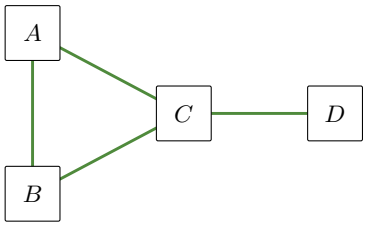
Level: Psychometrika. Observed variables: A, B, C, D ($p = 4$, 10 moments).

Model A $q_A = 7$




```
C ~ A + B; D ~ C
```

Model B $q_B = 8$




```
network A B C D; A - B + C; B - C; C - D
```

Claim. $A \preceq B$, $B \not\preceq A$: **A is nested in B** (strictly).

Proof. (i) $A \preceq B$. In A, D is independent of $\{A, B\}$ given C (local Markov property), so $(\Sigma^{-1})_{AD} = (\Sigma^{-1})_{BD} = 0$: $\Sigma \in \mathcal{M}_B$.

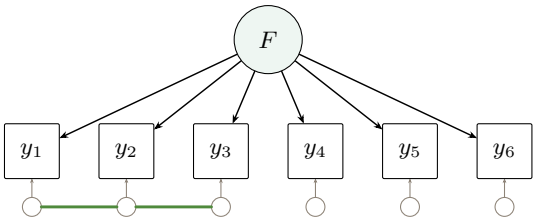
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 8 at the rational point listed in Appendix B, while model A has only $q_A = 7$ free parameters. □

In the game: B is the moral graph of the DAG in A (marry the parents of C). Every conditional independence in the GGM also holds in the DAG, but the DAG adds one: A and B are marginally independent.

P11 One residual edge inside a residual chain

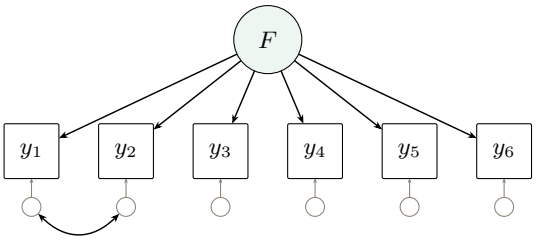
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3; y1 - y2; y2
- y3
```

Model B $q_B = 13$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 14 at the rational point listed in Appendix B, while model B has only $q_B = 13$ free parameters.

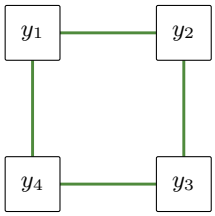
(ii) $B \preceq A$. By Lemma I the residual covariance θ_{12} alone is the residual network with only the edge $y_1 - y_2$; that is A with the edge $y_2 - y_3$ fixed to 0 (Lemma R). □

In the game: One residual covariance is one residual edge. A residual network with edges y1 — y2 — y3 contains the network with only y1 — y2, so B is nested in A.

P12 A chain inside a cycle network

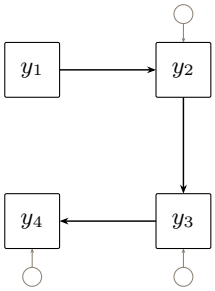
Level: Psychometrika. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4; y4 - y1
```

Model B $q_B = 7$




```
y2 ~ y1; y3 ~ y2; y4 ~ y3
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 8 at the rational point listed in Appendix B, while model B has only $q_B = 7$ free parameters.

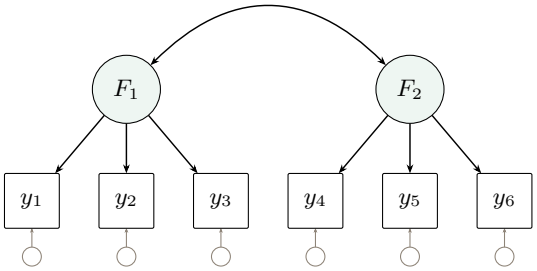
(ii) $B \preceq A$. By Lemma C, B is the GGM on the path $y_1 - y_2 - y_3 - y_4$, which is A with the edge $y_4 - y_1$ fixed to 0 (Lemma R). □

In the game: The directed chain equals the GGM with the path y1 — y2 — y3 — y4. The cycle network adds the edge y4 — y1, so it contains the chain.

P13 A tree network inside two factors

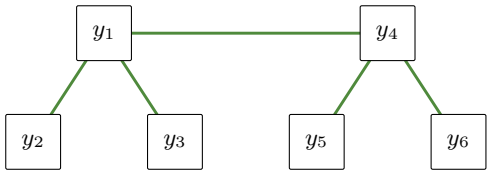
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 13$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F1 ~~ F2
```

Model B $q_B = 11$




```
network y1 y2 y3 y4 y5 y6; y1 - y2 + y3 + y4; y4 - y5 + y6
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 13 at the rational point listed in Appendix B, while model B has only $q_B = 11$ free parameters.

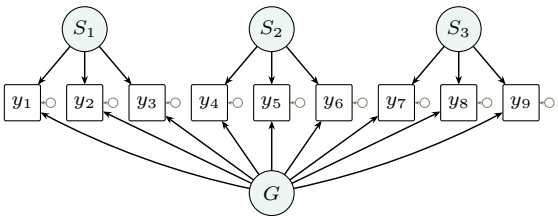
(ii) $B \preceq A$. By Lemma C, B equals the DAG $y_1 \to y_2, y_3, y_4$ and $y_4 \to y_5, y_6$. In A put $\theta_1 = \theta_4 = 0$, so $F_1 = y_1$ and $F_2 = y_4$ (Lemma T); then A reproduces this DAG ($\lambda_k = \beta_{k\cdot}$, $\psi_{12} = \mathrm{Cov}(y_1, y_4)$). □

In the game: Give y1 and y4 zero residual variance and they become F1 and F2 themselves. The two-factor model then becomes this tree-shaped network, so B is nested in A.

P14 A latent network inside a bifactor model

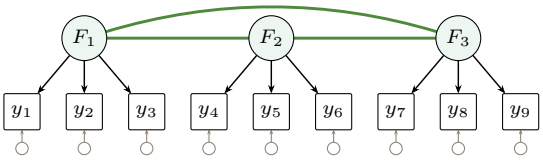
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9$ ($p = 9$, 45 moments).

Model A $q_A = 27$




```
S1 =~ y1 + y2 + y3; S2 =~ y4 + y5 + y6; S3 =~ y7 + y8 + y9; G
  =~ y1 + y2 + y3 + y4 + y5 + y6 + y7 + y8 + y9
```

Model B $q_B = 21$




```
F1 =~ y1 + y2 + y3; F2 =~ y4 + y5 + y6; F3 =~ y7 + y8 + y9;
network F1 F2 F3; F1 - F2 + F3; F2 - F3
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 27 at the rational point listed in Appendix B, while model B has only $q_B = 21$ free parameters.

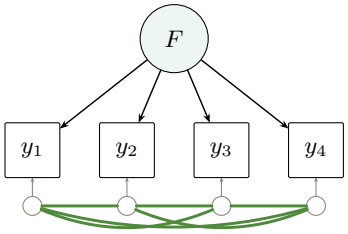
(ii) $B \preceq A$. A complete latent network has $\Psi = K^{-1}$ with K free, i.e. any $\Psi \succ 0$: B is the correlated three-factor model, which is nested in A by H07. □

In the game: A fully connected latent network is just three correlated factors, and those are nested in the bifactor model (via the higher-order model and Schmid–Leiman).

P15 Factor plus complete residual network is saturated

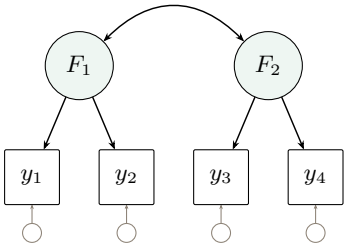
Level: Psychometrika. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4; network y1 y2 y3 y4; y1 - y2 + y3 + y4;
y2 - y3 + y4; y3 - y4
```

Model B $q_B = 9$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F1 ~~ F2
```

Claim. $B \preceq A$, $A \not\preceq B$: **B is nested in A** (strictly).

Proof. (i) $A \not\preceq B$ by Lemma D: the Jacobian of model A has rank 10 at the rational point listed in Appendix B, while model B has only $q_B = 9$ free parameters.

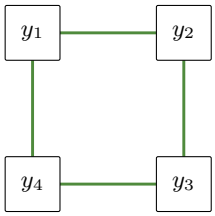
(ii) $B \preceq A$. Given any $\Sigma \succ 0$, choose λ and a small $\psi > 0$ with $\Sigma - \lambda\lambda^\top\psi \succ 0$, and $K = (\Sigma - \lambda\lambda^\top\psi)^{-1}$ (all edges free). So A is saturated and every model is nested in it. □

In the game: A factor model whose residuals form a complete network is saturated (and not identified): it reproduces any covariance matrix. Every model is nested in it.

P16 Cycle network versus one factor

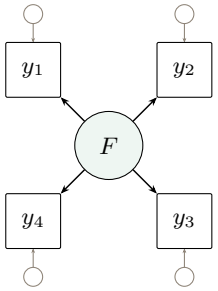
Level: Psychometrika. Observed variables: y_1, y_2, y_3, y_4 ($p = 4$, 10 moments).

Model A $q_A = 8$




```
network y1 y2 y3 y4; y1 - y2; y2 - y3; y3 - y4; y4 - y1
```

Model B $q_B = 8$




```
F =~ y1 + y2 + y3 + y4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{34} - \sigma_{13}\sigma_{24}$. A tetrad: zero in every one-factor model. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.006844 \neq 0$.

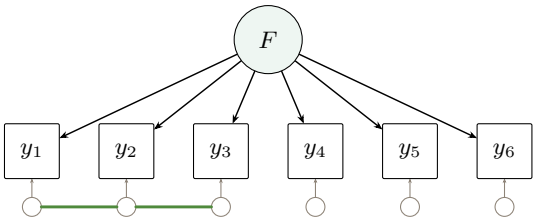
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{42} & \sigma_{43} & \sigma_{44} \end{pmatrix}$. This minor is the cofactor of $(\Sigma^{-1})_{13}$; in A, $k_{13} = 0$. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.8640 \neq 0$. □

In the game: The cycle network sets two partial correlations to zero (y1 — y3, y2 — y4); a factor model sets no partial correlation to zero but imposes tetrads. Both have 8 parameters, but the restrictions differ.

P17 Residual chain network versus residual covariance chain

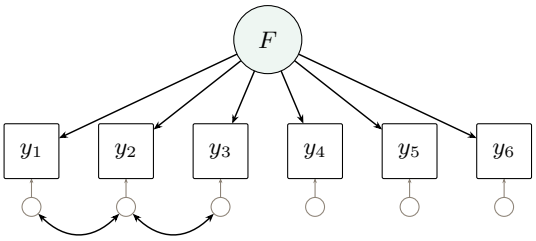
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6$ ($p = 6$, 21 moments).

Model A $q_A = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; network y1 y2 y3; y1 - y2; y2
- y3
```

Model B $q_B = 14$




```
F =~ y1 + y2 + y3 + y4 + y5 + y6; y1 ~~ y2; y2 ~~ y3
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{45} - \sigma_{14}\sigma_{35}$. In B, $\theta_{13} = 0$, so $\sigma_{13} = \lambda_1\lambda_3\psi$ and the tetrad vanishes. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \theta_{13}\lambda_4\lambda_5\psi \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.01863 \neq 0$.

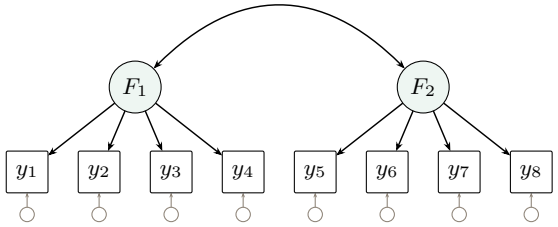
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = (\sigma_{13}\sigma_{45} - \sigma_{14}\sigma_{35})(\sigma_{22}\sigma_{45} - \sigma_{24}\sigma_{25}) - (\sigma_{12}\sigma_{45} - \sigma_{14}\sigma_{25})(\sigma_{23}\sigma_{45} - \sigma_{24}\sigma_{35})$. In A, $\sigma_{ij}\sigma_{45} - \sigma_{i4}\sigma_{j5} = \theta_{ij}\sigma_{45}$ for $i, j \in \{1, 2, 3\}$, so $f = \sigma_{45}^2(\theta_{13}\theta_{22} - \theta_{12}\theta_{23})$, and $k_{13} = 0$ means $\theta_{13}\theta_{22} = \theta_{12}\theta_{23}$ (Lemma I). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.05530 \neq 0$. □

In the game: A residual chain network sets a residual partial correlation to zero (y1, y3); the residual covariances in B set the residual covariance itself to zero. Same picture, different restriction.

P18 Correlated factors versus network cliques

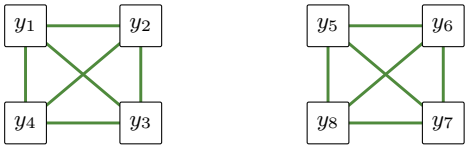
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 17$




```
F1 =~ y1 + y2 + y3 + y4; F2 =~ y5 + y6 + y7 + y8; F1 ~~ F2
```

Model B $q_B = 20$




```
network y1 y2 y3 y4 y5 y6 y7 y8; y1 - y2 + y3 + y4; y2 - y3 +
y4; y3 - y4; y5 - y6 + y7 + y8; y6 - y7 + y8; y7 - y8
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{15}$. In B there are no edges between the clusters, so Σ is block-diagonal (Lemma I). So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A, $f = \lambda_1\lambda_5\psi_{12} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = 0.3000 \neq 0$.

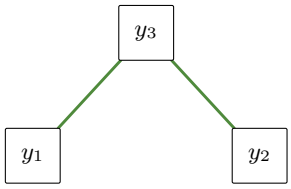
(ii) $B \not\preceq A$ by Lemma D: the Jacobian of model B has rank 20 at the rational point listed in Appendix B, while model A has only $q_A = 17$ free parameters. □

In the game: Correlated factors connect the clusters (the GGM cannot); the GGM leaves each cluster saturated (the factors impose tetrads). Neither is nested.

P19 Undirected path versus collider

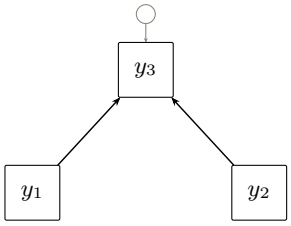
Level: Psychometrika. Observed variables: y_1, y_2, y_3 ($p = 3$, 6 moments).

Model A $q_A = 5$




```
network y1 y2 y3; y1 - y3; y3 - y2
```

Model B $q_B = 5$




```
y3 ~ y1 + y2
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{12}$. In B, y_1 and y_2 are exogenous and unconnected. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.01929 \neq 0$.

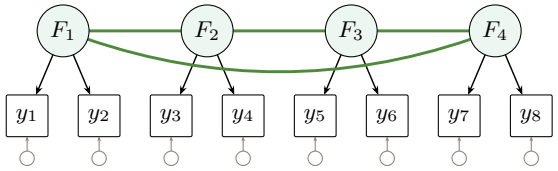
(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \sigma_{12}\sigma_{33} - \sigma_{13}\sigma_{23}$. In A, $k_{12} = 0$ (Lemma I). So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B, $f = -\sigma_{13}\sigma_{23} \neq 0$ for generic parameters; e.g. at the point of Appendix B, $f = -0.3600 \neq 0$. □

In the game: The network says y1 and y2 are independent given y3; the collider says they are independent without conditioning (and dependent given y3).

P20 Latent cycle network versus general factor

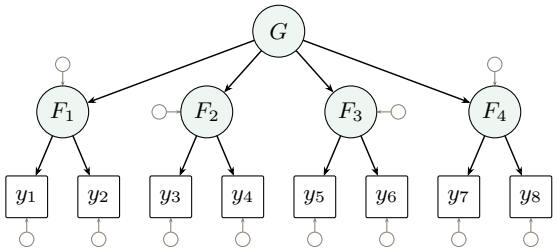
Level: Psychometrika. Observed variables: $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ($p = 8$, 36 moments).

Model A $q_A = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8;
network F1 F2 F3 F4; F1 - F2; F2 - F3; F3 - F4; F4 - F1
```

Model B $q_B = 20$




```
F1 =~ y1 + y2; F2 =~ y3 + y4; F3 =~ y5 + y6; F4 =~ y7 + y8; G
=~ F1 + F2 + F3 + F4
```

Claim. $A \not\preceq B$ and $B \not\preceq A$: **neither** is nested in the other.

Proof. (i) $A \not\preceq B$ by Lemma S with $f(\Sigma) = \sigma_{13}\sigma_{57} - \sigma_{15}\sigma_{37}$. In B the latent covariances follow one second-order factor: $\psi_{12}\psi_{34} = \psi_{13}\psi_{24}$. So $f \equiv 0$ on $\mathcal{M}_B$, whereas in model A at the point of Appendix B, $f = 0.006844 \neq 0$.

(ii) $B \not\preceq A$ by Lemma S with $f(\Sigma) = \det \begin{pmatrix} \sigma_{14} & \sigma_{16} & \sigma_{18} \\ \sigma_{34} & \sigma_{36} & \sigma_{38} \\ \sigma_{74} & \sigma_{76} & \sigma_{78} \end{pmatrix}$. The matrix C of covariances between first indicators (y_1, y_3, y_5, y_7) and second indicators (y_2, y_4, y_6, y_8) equals $D_1 \Psi D_2$ (D_k diagonal loadings), so $(C^{-1})_{13} \propto (\Psi^{-1})_{13} = k_{13} = 0$ in A; f is the corresponding cofactor. So $f \equiv 0$ on $\mathcal{M}_A$, whereas in model B at the point of Appendix B, $f = -0.6159 \neq 0$. □

In the game: Both explain the latent correlations with 8 parameters, but the cycle network zeroes two latent partial correlations while the general factor imposes two tetrads. Different restrictions, so neither is nested.

Appendix B Rational points used in Lemmas S and D

For every non-nesting argument: the model, the rational parameter values (in the order of the model's free parameters; $X \to Y$ is a loading or regression, Var a (residual) variance, Cov a (residual) covariance, k an element of a precision matrix) and the exact value of f or the exact rank of the Jacobian at that point. The implied Σ is positive definite at each point.

E06, model B (for $B \not\preceq A$): rank of the Jacobian $= 14 > q_A = 13$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

E07, model B (for $B \not\preceq A$): rank of the Jacobian $= 13 > q_A = 12$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

E08, model B (for $B \not\preceq A$): rank of the Jacobian $= 6 > q_A = 5$. $A \to B = \frac{1}{2}$, $A \to C = \frac{3}{5}$, $B \to C = \frac{2}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E09, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

E10, model B (for $B \not\preceq A$): rank of the Jacobian $= 13 > q_A = 12$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

E11, model A (for $A \not\preceq B$): rank of the Jacobian $= 14 > q_B = 13$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$.

E12, model A (for $A \not\preceq B$): rank of the Jacobian $= 6 > q_B = 5$. $A \to C = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Cov}(B, A) = \frac{3}{10}$.

E13, model A (for $A \not\preceq B$): rank of the Jacobian $= 8 > q_B = 6$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

E14, model A (for $A \not\preceq B$): rank of the Jacobian $= 9 > q_B = 8$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $X \to F = \frac{1}{2}$, $X \to y_1 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

E15, model A (for $A \not\preceq B$): rank of the Jacobian $= 5 > q_B = 3$. $A \to B = \frac{1}{2}$, $A \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E16, model A (for $A \not\preceq B$): $f = 0.300000$. $A \to B = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E16, model B (for $B \not\preceq A$): $f = -0.270000$. $A \to B = \frac{1}{2}$, $C \to B = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E17, model A (for $A \not\preceq B$): $f = 0.0648000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_2, y_1) = \frac{1}{5}$.

E17, model B (for $B \not\preceq A$): $f = 0.0600000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_6, y_5) = \frac{1}{5}$.

E18, model A (for $A \not\preceq B$): $f = 1.44000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

E18, model B (for $B \not\preceq A$): $f = 1.20000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$.

E19, model A (for $A \not\preceq B$): $f = 0.450000$. $A \to B = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E19, model B (for $B \not\preceq A$): $f = 0.600000$. $A \to C = \frac{1}{2}$, $C \to B = \frac{3}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$.

E20, model A (for $A \not\preceq B$): $f = 0.500000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_2 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$.

E20, model B (for $B \not\preceq A$): $f = 0.540000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_3 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$.

M06, model B (for $B \not\preceq A$): rank of the Jacobian $= 22 > q_A = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

M07, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 13$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_5, y_4) = \frac{3}{10}$, $\mathrm{Cov}(y_6, y_4) = \frac{1}{5}$, $\mathrm{Cov}(y_6, y_5) = \frac{2}{5}$.

M08, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 13$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

M09, model B (for $B \not\preceq A$): rank of the Jacobian $= 10 > q_A = 7$. $A \to B = \frac{1}{2}$, $A \to C = \frac{3}{5}$, $B \to C = \frac{2}{5}$, $A \to D = \frac{7}{10}$, $B \to D = \frac{3}{10}$, $C \to D = \frac{9}{20}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Var}(D) = \frac{4}{5}$.

M10, model B (for $B \not\preceq A$): rank of the Jacobian $= 13 > q_A = 12$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M11, model A (for $A \not\preceq B$): rank of the Jacobian $= 10 > q_B = 8$. $y_1 \to y_2 = \frac{1}{2}$, $y_1 \to y_3 = \frac{3}{5}$, $y_2 \to y_3 = \frac{2}{5}$, $y_1 \to y_4 = \frac{7}{10}$, $y_2 \to y_4 = \frac{3}{10}$, $y_3 \to y_4 = \frac{9}{20}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$.

M12, model A (for $A \not\preceq B$): rank of the Jacobian $= 11 > q_B = 10$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $X \to F = \frac{1}{2}$, $F \to Y = \frac{6}{5}$, $X \to Y = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(Y) = \frac{11}{10}$, $\mathrm{Var}(F) = 1$.

M13, model A (for $A \not\preceq B$): rank of the Jacobian $= 15 > q_B = 13$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

M14, model A (for $A \not\preceq B$): rank of the Jacobian $= 11 > q_B = 10$. $A \to C = \frac{1}{2}$, $B \to C = \frac{3}{5}$, $C \to D = \frac{2}{5}$, $C \to E = \frac{7}{10}$, $D \to E = \frac{3}{10}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Var}(D) = \frac{4}{5}$, $\mathrm{Var}(E) = \frac{11}{10}$, $\mathrm{Cov}(B, A) = \frac{3}{10}$.

M15, model A (for $A \not\preceq B$): rank of the Jacobian $= 6 > q_B = 5$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(F) = 1$.

M16, model A (for $A \not\preceq B$): $f = 1.17216$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M16, model B (for $B \not\preceq A$): $f = 0.888000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M17, model A (for $A \not\preceq B$): $f = 1.19880$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M17, model B (for $B \not\preceq A$): $f = 0.324000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$.

M18, model A (for $A \not\preceq B$): $f = 0.0720000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$.

M18, model B (for $B \not\preceq A$): $f = 1.17216$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_5 = \frac{4}{5}$, $F_2 \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

M19, model A (for $A \not\preceq B$): rank of the Jacobian $= 8 > q_B = 7$. $X \to M_1 = \frac{1}{2}$, $X \to M_2 = \frac{3}{5}$, $M_1 \to Y = \frac{2}{5}$, $M_2 \to Y = \frac{7}{10}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M_1) = \frac{6}{5}$, $\mathrm{Var}(M_2) = \frac{9}{10}$, $\mathrm{Var}(Y) = \frac{4}{5}$.

M19, model B (for $B \not\preceq A$): $f = 0.720000$. $X \to M_1 = \frac{1}{2}$, $M_1 \to M_2 = \frac{3}{5}$, $M_2 \to Y = \frac{2}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(M_1) = \frac{6}{5}$, $\mathrm{Var}(M_2) = \frac{9}{10}$, $\mathrm{Var}(Y) = \frac{4}{5}$.

M20, model A (for $A \not\preceq B$): $f = 0.324000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$.

M20, model B (for $B \not\preceq A$): $f = 0.360000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_3, y_1) = \frac{3}{10}$.

H06, model B (for $B \not\preceq A$): rank of the Jacobian $= 27 > q_A = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H07, model B (for $B \not\preceq A$): rank of the Jacobian $= 27 > q_A = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H08, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_1 \to F_2 = \frac{1}{2}$, $F_1 \to F_3 = \frac{3}{5}$, $F_2 \to F_3 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$.

H09, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $F \to B = 1$, $F \to C = \frac{9}{10}$, $F \to D = \frac{6}{5}$, $\mathrm{Var}(A) = 1$, $\mathrm{Var}(B) = \frac{6}{5}$, $\mathrm{Var}(C) = \frac{9}{10}$, $\mathrm{Var}(D) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

H10, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

H11, model A (for $A \not\preceq B$): rank of the Jacobian $= 22 > q_B = 20$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

H12, model A (for $A \not\preceq B$): rank of the Jacobian $= 9 > q_B = 8$. $F \to y_1 = 1$, $F \to y_2 = \frac{9}{10}$, $F \to y_3 = \frac{6}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_1, X) = \frac{3}{10}$.

H13, model A (for $A \not\preceq B$): rank of the Jacobian $= 13 > q_B = 12$. $X_1 \to X_3 = \frac{1}{2}$, $X_2 \to X_3 = \frac{3}{5}$, $X_1 \to X_4 = \frac{2}{5}$, $X_2 \to X_4 = \frac{7}{10}$, $X_3 \to X_4 = \frac{3}{10}$, $X_4 \to X_5 = \frac{9}{20}$, $X_2 \to X_5 = \frac{1}{2}$, $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = \frac{6}{5}$, $\mathrm{Var}(X_3) = \frac{9}{10}$, $\mathrm{Var}(X_4) = \frac{4}{5}$, $\mathrm{Var}(X_5) = \frac{11}{10}$, $\mathrm{Cov}(X_2, X_1) = \frac{3}{10}$.

H14, model A (for $A \not\preceq B$): rank of the Jacobian $= 15 > q_B = 13$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_4, y_3) = \frac{1}{5}$, $\mathrm{Cov}(y_6, y_5) = \frac{2}{5}$.

H15, model A (for $A \not\preceq B$): rank of the Jacobian $= 22 > q_B = 17$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_4, F_1) = \frac{2}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{1}{4}$, $\mathrm{Cov}(F_4, F_2) = \frac{7}{20}$, $\mathrm{Cov}(F_4, F_3) = \frac{3}{20}$.

H16, model A (for $A \not\preceq B$): $f = 0.180000$. $y_1 \to y_2 = \frac{1}{2}$, $y_2 \to y_3 = \frac{3}{5}$, $y_3 \to y_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$.

H16, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

H17, model A (for $A \not\preceq B$): $f = 0.274725$. $X_1 \to X_3 = \frac{1}{2}$, $X_2 \to X_3 = \frac{3}{5}$, $X_1 \to X_4 = \frac{2}{5}$, $X_2 \to X_4 = \frac{7}{10}$, $X_3 \to X_4 = \frac{3}{10}$, $X_4 \to X_5 = \frac{9}{20}$, $X_2 \to X_5 = \frac{1}{2}$, $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = \frac{6}{5}$, $\mathrm{Var}(X_3) = \frac{9}{10}$, $\mathrm{Var}(X_4) = \frac{4}{5}$, $\mathrm{Var}(X_5) = \frac{11}{10}$, $\mathrm{Cov}(X_2, X_1) = \frac{3}{10}$.

H17, model B (for $B \not\preceq A$): $f = -0.122100$. $X_1 \to X_3 = \frac{1}{2}$, $X_2 \to X_3 = \frac{3}{5}$, $X_1 \to X_4 = \frac{2}{5}$, $X_2 \to X_4 = \frac{7}{10}$, $X_3 \to X_4 = \frac{3}{10}$, $X_2 \to X_5 = \frac{9}{20}$, $\mathrm{Var}(X_1) = 1$, $\mathrm{Var}(X_2) = \frac{6}{5}$, $\mathrm{Var}(X_3) = \frac{9}{10}$, $\mathrm{Var}(X_4) = \frac{4}{5}$, $\mathrm{Var}(X_5) = \frac{11}{10}$, $\mathrm{Cov}(X_2, X_1) = \frac{3}{10}$, $\mathrm{Cov}(X_5, X_4) = \frac{1}{5}$.

H18, model A (for $A \not\preceq B$): $f = 0.950400$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $F_3 \to y_8 = \frac{11}{10}$, $F_3 \to y_9 = \frac{7}{10}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

H18, model B (for $B \not\preceq A$): $f = 0.0840000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $F \to y_7 = \frac{7}{10}$, $F \to y_8 = \frac{13}{10}$, $F \to y_9 = \frac{3}{4}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_3, y_1) = \frac{1}{5}$, $\mathrm{Cov}(y_3, y_2) = \frac{2}{5}$.

H19, model A (for $A \not\preceq B$): $f = 0.540000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_1 = \frac{1}{2}$, $F_1 \to F_2 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

H19, model B (for $B \not\preceq A$): $f = 0.500000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $X \to F_2 = \frac{1}{2}$, $F_2 \to F_1 = \frac{3}{5}$, $\mathrm{Var}(X) = 1$, $\mathrm{Var}(y_1) = \frac{6}{5}$, $\mathrm{Var}(y_2) = \frac{9}{10}$, $\mathrm{Var}(y_3) = \frac{4}{5}$, $\mathrm{Var}(y_4) = \frac{11}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$.

H20, model A (for $A \not\preceq B$): rank of the Jacobian $= 20 > q_B = 19$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$.

H20, model B (for $B \not\preceq A$): $f = 0.160000$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $F_1 \to F_2 = \frac{1}{2}$, $F_2 \to F_3 = \frac{3}{5}$, $F_3 \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$.

P06, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

P07, model B (for $B \not\preceq A$): rank of the Jacobian $= 20 > q_A = 16$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_5,y_5} = \frac{12}{5}$, $k_{y_6,y_6} = \frac{21}{10}$, $k_{y_7,y_7} = 2$, $k_{y_8,y_8} = \frac{9}{4}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$, $k_{y_3,y_2} = -\frac{1}{4}$, $k_{y_4,y_2} = -\frac{7}{20}$, $k_{y_4,y_3} = -\frac{9}{20}$, $k_{y_6,y_5} = -\frac{1}{2}$, $k_{y_7,y_5} = -\frac{2}{5}$, $k_{y_8,y_5} = -\frac{3}{10}$, $k_{y_7,y_6} = -\frac{1}{4}$, $k_{y_8,y_6} = -\frac{7}{20}$, $k_{y_8,y_7} = -\frac{9}{20}$.

P08, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

P09, model B (for $B \not\preceq A$): rank of the Jacobian $= 15 > q_A = 14$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$, $\mathrm{Cov}(F_3, F_1) = \frac{1}{5}$, $\mathrm{Cov}(F_3, F_2) = \frac{2}{5}$.

P10, model B (for $B \not\preceq A$): rank of the Jacobian $= 8 > q_A = 7$. $k_{A,A} = 2$, $k_{B,B} = \frac{9}{4}$, $k_{C,C} = \frac{5}{2}$, $k_{D,D} = \frac{11}{5}$, $k_{B,A} = -\frac{1}{2}$, $k_{C,A} = -\frac{2}{5}$, $k_{C,B} = -\frac{3}{10}$, $k_{D,C} = -\frac{1}{4}$.

P11, model A (for $A \not\preceq B$): rank of the Jacobian $= 14 > q_B = 13$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $\mathrm{Var}(y_4) = 1$, $\mathrm{Var}(y_5) = \frac{6}{5}$, $\mathrm{Var}(y_6) = \frac{9}{10}$, $\mathrm{Var}(F) = 1$.

P12, model A (for $A \not\preceq B$): rank of the Jacobian $= 8 > q_B = 7$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_3} = -\frac{3}{10}$, $k_{y_4,y_1} = -\frac{1}{4}$.

P13, model A (for $A \not\preceq B$): rank of the Jacobian $= 13 > q_B = 11$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_2 \to y_5 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

P14, model A (for $A \not\preceq B$): rank of the Jacobian $= 27 > q_B = 21$. $S_1 \to y_2 = 1$, $S_1 \to y_3 = \frac{9}{10}$, $S_2 \to y_5 = \frac{6}{5}$, $S_2 \to y_6 = \frac{4}{5}$, $S_3 \to y_8 = \frac{11}{10}$, $S_3 \to y_9 = \frac{7}{10}$, $G \to y_2 = \frac{13}{10}$, $G \to y_3 = \frac{3}{4}$, $G \to y_4 = \frac{17}{20}$, $G \to y_5 = \frac{21}{20}$, $G \to y_6 = 1$, $G \to y_7 = \frac{9}{10}$, $G \to y_8 = \frac{6}{5}$, $G \to y_9 = \frac{4}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(y_9) = \frac{19}{20}$, $\mathrm{Var}(S_1) = 1$, $\mathrm{Var}(S_2) = \frac{6}{5}$, $\mathrm{Var}(S_3) = \frac{4}{5}$, $\mathrm{Var}(G) = \frac{11}{10}$.

P15, model A (for $A \not\preceq B$): rank of the Jacobian $= 10 > q_B = 9$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$, $k_{y_3,y_2} = -\frac{1}{4}$, $k_{y_4,y_2} = -\frac{7}{20}$, $k_{y_4,y_3} = -\frac{9}{20}$, $\mathrm{Var}(F) = 1$.

P16, model A (for $A \not\preceq B$): $f = 0.00684402$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $k_{y_4,y_3} = -\frac{3}{10}$, $k_{y_4,y_1} = -\frac{1}{4}$.

P16, model B (for $B \not\preceq A$): $f = -0.864000$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(F) = 1$.

P17, model A (for $A \not\preceq B$): $f = 0.0186317$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$, $\mathrm{Var}(y_4) = 1$, $\mathrm{Var}(y_5) = \frac{6}{5}$, $\mathrm{Var}(y_6) = \frac{9}{10}$, $\mathrm{Var}(F) = 1$.

P17, model B (for $B \not\preceq A$): $f = -0.0552960$. $F \to y_2 = 1$, $F \to y_3 = \frac{9}{10}$, $F \to y_4 = \frac{6}{5}$, $F \to y_5 = \frac{4}{5}$, $F \to y_6 = \frac{11}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(F) = 1$, $\mathrm{Cov}(y_2, y_1) = \frac{3}{10}$, $\mathrm{Cov}(y_3, y_2) = \frac{1}{5}$.

P18, model A (for $A \not\preceq B$): $f = 0.300000$. $F_1 \to y_2 = 1$, $F_1 \to y_3 = \frac{9}{10}$, $F_1 \to y_4 = \frac{6}{5}$, $F_2 \to y_6 = \frac{4}{5}$, $F_2 \to y_7 = \frac{11}{10}$, $F_2 \to y_8 = \frac{7}{10}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Cov}(F_2, F_1) = \frac{3}{10}$.

P18, model B (for $B \not\preceq A$): rank of the Jacobian $= 20 > q_A = 17$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_4,y_4} = \frac{11}{5}$, $k_{y_5,y_5} = \frac{12}{5}$, $k_{y_6,y_6} = \frac{21}{10}$, $k_{y_7,y_7} = 2$, $k_{y_8,y_8} = \frac{9}{4}$, $k_{y_2,y_1} = -\frac{1}{2}$, $k_{y_3,y_1} = -\frac{2}{5}$, $k_{y_4,y_1} = -\frac{3}{10}$, $k_{y_3,y_2} = -\frac{1}{4}$, $k_{y_4,y_2} = -\frac{7}{20}$, $k_{y_4,y_3} = -\frac{9}{20}$, $k_{y_6,y_5} = -\frac{1}{2}$, $k_{y_7,y_5} = -\frac{2}{5}$, $k_{y_8,y_5} = -\frac{3}{10}$, $k_{y_7,y_6} = -\frac{1}{4}$, $k_{y_8,y_6} = -\frac{7}{20}$, $k_{y_8,y_7} = -\frac{9}{20}$.

P19, model A (for $A \not\preceq B$): $f = 0.0192911$. $k_{y_1,y_1} = 2$, $k_{y_2,y_2} = \frac{9}{4}$, $k_{y_3,y_3} = \frac{5}{2}$, $k_{y_3,y_1} = -\frac{1}{2}$, $k_{y_3,y_2} = -\frac{2}{5}$.

P19, model B (for $B \not\preceq A$): $f = -0.360000$. $y_1 \to y_3 = \frac{1}{2}$, $y_2 \to y_3 = \frac{3}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$.

P20, model A (for $A \not\preceq B$): $f = 0.00684402$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $k_{F_1,F_1} = 2$, $k_{F_2,F_2} = \frac{9}{4}$, $k_{F_3,F_3} = \frac{5}{2}$, $k_{F_4,F_4} = \frac{11}{5}$, $k_{F_2,F_1} = -\frac{1}{2}$, $k_{F_3,F_2} = -\frac{2}{5}$, $k_{F_4,F_3} = -\frac{3}{10}$, $k_{F_4,F_1} = -\frac{1}{4}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$.

P20, model B (for $B \not\preceq A$): $f = -0.615859$. $F_1 \to y_2 = 1$, $F_2 \to y_4 = \frac{9}{10}$, $F_3 \to y_6 = \frac{6}{5}$, $F_4 \to y_8 = \frac{4}{5}$, $G \to F_2 = \frac{1}{2}$, $G \to F_3 = \frac{3}{5}$, $G \to F_4 = \frac{2}{5}$, $\mathrm{Var}(y_1) = 1$, $\mathrm{Var}(y_2) = \frac{6}{5}$, $\mathrm{Var}(y_3) = \frac{9}{10}$, $\mathrm{Var}(y_4) = \frac{4}{5}$, $\mathrm{Var}(y_5) = \frac{11}{10}$, $\mathrm{Var}(y_6) = \frac{7}{10}$, $\mathrm{Var}(y_7) = \frac{13}{10}$, $\mathrm{Var}(y_8) = \frac{3}{4}$, $\mathrm{Var}(F_1) = 1$, $\mathrm{Var}(F_2) = \frac{6}{5}$, $\mathrm{Var}(F_3) = \frac{4}{5}$, $\mathrm{Var}(F_4) = \frac{11}{10}$, $\mathrm{Var}(G) = \frac{9}{10}$.

References

Bentler, P. M., & Satorra, A. (2010). Testing model nesting and equivalence. *Psychological Methods, 15*, 111–123.

Epskamp, S., Rhemtulla, M., & Borsboom, D. (2017). Generalized network psychometrics: Combining network and latent variable models. *Psychometrika, 82*, 904–927.

Lauritzen, S. L. (1996). *Graphical models*. Oxford: Clarendon Press.

Lee, S., & Hershberger, S. (1990). A simple rule for generating equivalent models in covariance structure modeling. *Multivariate Behavioral Research, 25*, 313–334.

MacCallum, R. C., Wegener, D. T., Uchino, B. N., & Fabrigar, L. R. (1993). The problem of equivalent models in applications of covariance structure analysis. *Psychological Bulletin, 114*, 185–199.

Schmid, J., & Leiman, J. M. (1957). The development of hierarchical factor solutions. *Psychometrika, 22*, 53–61.

Yung, Y.-F., Thissen, D., & McLeod, L. D. (1999). On the relationship between the higher-order factor model and the hierarchical factor model. *Psychometrika, 64*, 113–128.